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Title

AI-Powered Diagnosis for Macular OCT Imaging

Defended publicly on 13/06/2026 in front of the jury composed of:

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Author Declaration

I, **Hafida Boukedjar**, declare that the work presented in this dissertation, "**AI-Powered Diagnosis for Macular OCT Imaging**", is original and authored by me, except where otherwise indicated through formal citation.

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24 May 2026

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Besmala Bendif

24 May 2026

AI Usage Disclosure

We hereby disclose the involvement of Large Language Models (LLMs) and AI-driven tools in the preparation of this dissertation. While these tools facilitated technical optimization and linguistic refinement, We maintain full accountability for the accuracy, intellectual integrity, and originality of the research presented.

The following AI tools were utilized as follows:

- **Conceptualization and Code:** ChatGPT-4 and ChatGPT-5 were employed to assist with code debugging, implementation guidance. Perplexity was occasionally consulted for additional technical information and support.
- **Data Processing:** ChatGPT-4 and ChatGPT-5 helped with understanding acronyms and supporting data preprocessing and organization tasks.
- **Visual Content:** Figures and workflow illustrations were generated using ChatGPT-4 and ChatGPT-5 to enhance clarity and presentation.
- **Manuscript Preparation:** ChatGPT and Claude were used to improve writing quality, grammar, and structure of the manuscript.

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Hafida Boukedjar



Besmala Bnedif

Dedication

Hafida Boukedjar

This work is dedicated to my family for their continuous support, encouragement, and guidance throughout my academic journey.

To my father, **Zineddine Boukedjar**, and my mother, **Wafaa Kheddami**, whose sacrifices, patience, and unwavering support have been essential to my academic achievement.

Their constant encouragement, even from a distance, has been a fundamental source of strength throughout this journey.

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I also dedicate this work to the memory of my uncle, **Nedjmeddine Kheddami**, whose support and encouragement continue to inspire me even after his passing during the course of this work.

I express my sincere gratitude to my family for their constant belief in me and their unwavering support throughout the realization of this work.

I also dedicate this achievement to myself, in recognition of the perseverance, discipline, and commitment required throughout these years.

Special thanks are extended to my binome, **Besmala Bendif**, for her collaboration and support throughout this project.

Finally, I would like to express my sincere gratitude to our supervisors, for their guidance, valuable advice, and continuous support throughout this work.

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الملخص

تُعد أمراض الشبكية والبقعة الصفراء، بما في ذلك التنكس البقي المرتبط بالعمر، والوذمة البقعية الناتجة عن داء السكري، واضطرابات الأوعية الدموية الشبكية، من بين الأسباب الرئيسية لفقدان البصر غير القابل للعكس على مستوى العالم. ويُعد التشخيص المبكر أمراً أساسياً للحفاظ على القدرة البصرية، إلا أن التفسير اليدوي لفحوصات التصوير المقطعي للتماسك البصري لا يزال يستغرق وقتاً طويلاً ويصعب توسيع نطاقه في الممارسة السريرية الروتينية. وعلى الرغم من أن مناهج التعلم العميق الحديثة قد أظهرت نتائج واعدة، فإن العديد من الأنظمة الحالية تعتمد فقط على بيانات الصور وتفتقر إلى قابلية التفسير والتحقق السريري الواقعي الكافي.

تقترح هذه الأطروحة إطاراً قائماً على الذكاء الاصطناعي لتشخيص أمراض الشبكية اعتماداً على صور التصوير المقطعي للتماسك البصري من خلال مساهمتين متكاملتين. تقدم المساهمة الأولى إطاراً للتعلم متعدد الوسائط باستخدام مجموعة البيانات المرجعية العامة الخاصة بتشخيص أمراض الشبكية. وتم إجراء دراسة منهجية لاستبعاد المكونات وتقييم مشفرات الصور واستراتيجيات ترميز البيانات الوصفية وآليات الدمج المختلفة. يجمع النموذج النهائي بين مشفر بصري قائم على المحول الهرمي للصور وبيانات سريرية منظمة باستخدام آلية دمج تعتمد على الانتباه المتبادل ثنائي الاتجاه، مما يسمح بالتفاعل بين المؤشرات الحيوية للشبكية والمعلومات السريرية الخاصة بالمريض. أظهرت النتائج التجريبية دقة اختبار بلغت 30.93 %، ودرجة F1 موزونة بلغت 91.92 %، ودرجة F1 كلية بلغت 52.90 %.

تقدم المساهمة الثانية إطاراً لدعم القرار موجهاً سريرياً تم تطويره باستخدام مجموعة بيانات واقعية للتصوير المقطعي للتماسك البصري جُمعت من مركز ابن الهيثم لطب العيون. ولمعالجة مشكلة الإشراف الضعيف والتباين بين المرضى، تم اقتراح خط أنابيب تشخيص هرمي يعتمد على التعلم متعدد الحالات باستخدام آلية الانتباه الموجه لتحليل صور التصوير المقطعي الحجمية.

ولتعزيز الثقة السريرية، تم دمج خرائط تنشيط المناطق المهمة، والتدرجات المتكاملة، وإسناد طبقات الشبكية الموجه تشريحياً لتوفير قابلية التفسير. وتم نشر النماذج المقترحة من خلال منصة ذكاء الشبكية، وهي منصة سريرية آمنة تدعم إدارة المرضى، والتحكم في صلاحيات الوصول، والتشخيص العيني المدعوم بالذكاء الاصطناعي. الكلمات المفتاحية: التصوير المقطعي للتماسك البصري، تشخيص أمراض الشبكية، التعلم العميق متعدد الوسائط، دمج الانتباه المتبادل، قابلية التفسير، أنظمة دعم القرار السريري.

Abstract

Retinal and macular diseases are among the leading causes of irreversible visual impairment worldwide, where early diagnosis is essential for preserving vision. However, manual interpretation of Optical Coherence Tomography (OCT) remains time-consuming and difficult to scale in clinical practice. While deep learning approaches have shown promising results, many rely only on imaging data.

This work proposes an AI-driven framework for OCT-based retinal disease diagnosis through two complementary contributions. The first introduces a multimodal learning framework evaluated on the OCTDL benchmark, where multiple image encoders, structured metadata representations, and fusion strategies are systematically investigated to identify the optimal diagnostic architecture. The framework integrates retinal OCT imaging with structured clinical metadata, including patient age, sex, and eye laterality, enabling complementary information to be incorporated into the diagnostic process. The final model combines a Swin Transformer-Base visual encoder with an MLP-based metadata encoder using dual cross-attention fusion, achieving 93.50% accuracy, 92.91% weighted F1-score, and 90.52% macro F1-score.

The second contribution presents a clinical decision support system designed for real-world deployment using a private patient-level OCT dataset collected under weak supervision. A hierarchical two-stage framework based on Gated Attention Multiple Instance Learning (GA-MIL) was developed for volumetric patient-level diagnosis. Stage 1 performs binary screening to distinguish normal and pathological cases, achieving an accuracy of 96.88%, an F1-score of 98.21%, and a ROC-AUC of 98.75%. Stage 2 performs disease-specific classification for pathological cases and achieved a macro ROC-AUC of

79%, demonstrating the effectiveness of the proposed framework for robust patient-level diagnosis under real clinical conditions.

To enhance clinical trust, Grad-CAM, Integrated Gradients, and retinal layer attribution are used for explainability. The system is deployed through RetinAI, a secure clinical platform supporting patient management, access control, and AI-assisted diagnosis.

Keywords: Optical Coherence Tomography (OCT), retinal disease diagnosis, multi-modal deep learning, vision transformers, Swin Transformer, cross-attention fusion, multiple instance learning, patient-level diagnosis, Gated Attention, explainability, Grad-CAM, Integrated Gradients, clinical decision support systems, RetinAI.

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List of Acronyms

Acc Accuracy. [46](#), [48–51](#), [55](#), [57](#), [59](#), [60](#), [63](#), [66](#)

AMD Age-Related Macular Degeneration. [1](#), [7–10](#), [14](#), [19](#), [21](#), [22](#), [27](#), [43](#), [48](#), [57](#), [66](#), [67](#)

AMP Automatic Mixed Precision. [41](#), [44](#), [62](#)

B2B Business-to-Business. [75](#)

CAGR Compound Annual Growth Rate. [75](#), [78](#)

CAPEX Capital Expenditure. [81–83](#)

CDSS Clinical Decision Support System. [68](#)

CLAHE Contrast Limited Adaptive Histogram Equalization. [33](#)

CNN Convolutional Neural Network. [2](#)

CNV Choroidal Neovascularization. [7](#), [13](#), [19](#)

DCA Dual Cross-Attention. [xvi](#), [39](#), [40](#), [57](#)

DME Diabetic Macular Edema. [1](#), [8–10](#), [13](#), [19](#), [21](#), [22](#), [27](#)

DR Diabetic Retinopathy. [8](#), [43](#), [66](#)

ECE Expected Calibration Error. [28](#), [51](#), [52](#), [55](#), [63](#), [87](#)

ERM Epiretinal Membrane. [27](#), [48](#)

FMEA Failure Mode and Effects Analysis. [84](#)

GA Gated Attention. [5](#), [41](#), [42](#), [44](#), [66](#)

GA-MIL Gated Attention Multiple Instance Learning. [4](#), [5](#)

GCIPL Ganglion Cell-Inner Plexiform Layer. [10](#)

Grad-CAM Gradient-weighted Class Activation Mapping. [xv](#), [17](#), [29](#), [30](#), [55](#), [56](#), [58](#), [59](#),
[64–66](#), [70](#), [71](#), [77–79](#)

ILM Internal Limiting Membrane. [30](#)

INL Inner Nuclear Layer. [30](#)

IPL Inner Plexiform Layer. [30](#)

IT Information Technology. [76](#), [79](#)

LAN Local Area Network. [76](#)

MIL Multiple Instance Learning. [xv](#), [xvii](#), [5](#), [25–27](#), [34](#), [42](#), [44](#), [62](#), [64–67](#), [87](#)

MLP Multilayer Perceptron. [36](#), [37](#), [40](#), [49–51](#)

MVP Minimum Viable Product. [77](#)

NO Normal. [27](#), [48](#)

OCT Optical Coherence Tomography. [xiv](#), [xv](#), [xvii](#), [1–19](#), [21–36](#), [39–42](#), [44–46](#), [48–51](#),
[55–58](#), [60–63](#), [65–81](#), [86](#), [88](#), [89](#)

ONL Outer Nuclear Layer. [30](#)

OPEX Operating Expenditure. [81–83](#)

PR-AUC Precision-Recall Area Under the Curve. [46](#), [64](#), [66](#)

RAO Retinal Artery Occlusion. [27](#), [48](#)

RBAC Role-Based Access Control. [70](#), [71](#), [79](#), [84](#), [85](#)

RNFL Retinal Nerve Fiber Layer. [10](#), [30](#)

ROC-AUC Receiver Operating Characteristic - Area Under the Curve. [xvii](#), [28](#), [46](#),
[51–53](#), [61–64](#), [66](#)

ROI Region of Interest. [34](#)

RPE Retinal Pigment Epithelium. [7](#), [20](#), [30](#)

RPN Risk Priority Number. [84](#), [85](#)

RVO Retinal Vein Occlusion. [27](#), [48](#)

SaaS Software-as-a-Service. [75](#), [80](#), [85](#)

VID Vitreoretinal Interface Disorder. [27](#), [48](#)

ViT Vision Transformer. [xiv](#), [2](#), [14](#), [35–37](#), [49](#)

VMI Vitreomacular Interface. [1](#)

Genaral Introduction

0.1 General Context and Motivation

Artificial intelligence has emerged as a transformative paradigm in medical image analysis, enabling automated systems to achieve clinically competitive performance across radiology, pathology, and ophthalmology (Litjens et al., 2017). Within this domain, retinal imaging has attracted significant scientific interest due to the increasing global prevalence of vision-threatening diseases and the need for scalable ophthalmic screening solutions.

Retinal pathologies, including Age-Related Macular Degeneration (AMD), Diabetic Macular Edema (DME), and Vitreomacular Interface (VMI) abnormalities, are among the leading causes of irreversible visual impairment worldwide (Wong et al., 2014). Early detection is essential to prevent permanent vision loss; however, current clinical diagnosis relies heavily on manual interpretation of OCT examinations by specialists. This process is time-consuming, subject to inter-observer variability, and difficult to scale in resource limited healthcare systems.

OCT is considered the gold standard imaging modality in ophthalmology, providing non-invasive, high-resolution, cross-sectional visualization of retinal microstructure (Huang et al., 1991b). Despite its diagnostic value, interpretation of full OCT volumes remains challenging, as pathological patterns are often subtle, spatially heterogeneous, and distributed across multiple retinal layers. These challenges are further amplified in real-world settings due to variability in acquisition devices, imaging protocols, and patient populations.

Although deep learning approaches such as [Convolutional Neural Network \(CNN\)](#), [ViT](#), and multimodal architectures have significantly advanced [OCT](#)-based diagnosis ([De Fauw et al., 2018b](#)), most existing methods remain limited in three key aspects: (i) they rely solely on imaging data while ignoring clinical metadata, (ii) they process individual B-scans instead of full volumetric examinations, and (iii) they use opaque black-box models that limit interpretability and clinical trust ([Rudin, 2019](#)).

These limitations highlight the need for clinically oriented AI systems that integrate multimodal data, exploit volumetric information under weak supervision, and provide transparent and anatomically grounded explanations for reliable retinal disease diagnosis.

0.2 Problem Statement

Several persistent limitations hinder the clinical deployment of AI-based retinal diagnostic systems. First, most models are trained on curated public datasets that fail to capture the distributional variability of real clinical environments, leading to performance degradation upon deployment, commonly referred to as domain shift ([Guan and Liu, 2021](#)). Second, available patient-specific metadata such as age, sex, and eye laterality is often ignored. Moreover, when available, this information is frequently noisy, inconsistent, or incomplete, despite its established relevance to ophthalmic diagnosis. Third, processing individual [OCT](#) B-scans in isolation disregards the volumetric nature of the acquisition, thereby reducing clinical fidelity. Finally, the predominance of non-interpretable models raises substantive concerns regarding reliability, regulatory compliance, and patient safety ([Rudin, 2019](#)).

These limitations collectively underscore the need for clinically oriented AI frameworks capable of integrating multimodal data, performing patient-level volumetric reasoning under weak supervision, and providing transparent, anatomically grounded explanations.

In this context, this work addresses the following research questions:

- Does multimodal fusion of [OCT](#) imaging and clinical metadata yield statistically meaningful improvements over unimodal image-based classification?

- How can missing, noisy, or inconsistent patient metadata be recovered to support robust clinical AI modeling?
- Which fusion strategy offers the best balance between classification accuracy, robustness, and calibration?
- Can patient-level Gated Attention Multiple Instance Learning enhance diagnostic reliability under real-world weak supervision?
- Do gradient-based attribution methods produce clinically meaningful and anatomically coherent explanations of model decisions?
- How can AI-based retinal diagnosis frameworks be effectively translated into deployable clinical decision support tools?

0.3 Research Objectives and Contribution

The primary objective of this work is to develop and evaluate clinically reliable AI frameworks for OCT-based retinal disease diagnosis under both benchmark and real-world clinical conditions. The specific objectives are:

- Develop a reliable preprocessing framework that improves data quality and enables robust integration into multimodal clinical AI systems.
- Develop a multimodal deep learning framework integrating OCT imaging with structured clinical metadata for retinal disease classification.
- Conduct systematic ablation studies across transformer-based visual encoders, metadata encoding strategies, and multimodal fusion mechanisms.
- Design a hierarchical, patient-level diagnostic pipeline based on weakly supervised Multiple Instance Learning for volumetric OCT analysis.
- Integrate gradient-based and attribution explainability techniques to provide clinically interpretable and anatomically grounded predictions.

- Evaluate model performance, calibration, and generalization on both public benchmark and real-world private clinical datasets.
- Deploy the proposed frameworks within RETINAAI, a clinical decision support platform supporting AI-assisted ophthalmic diagnosis and secure patient management.

Contribution

This work proposes two complementary AI frameworks for OCT-based retinal disease analysis.

The **Framework I** focuses on multimodal retinal disease classification using the publicly available OCTDL benchmark (Kulyabin et al., 2024). The proposed architecture combines transformer-based visual encoding of OCT images with structured clinical metadata through a dual cross-attention fusion mechanism, enabling effective multimodal interaction and improved diagnostic performance.

The **Framework II** targets real-world clinical deployment using a private OCT dataset collected from routine ophthalmic examinations. It is based on a hierarchical Gated Attention Multiple Instance Learning (GA-MIL) (Ilse et al., 2018) pipeline that performs binary abnormality screening followed by disease-specific multiclass classification at the patient-examination level under weak supervision using only examination-level labels.

In addition, this work introduces a metadata recovery module designed to handle missing, noisy, and inconsistent patient information, thereby improving data completeness and enabling reliable integration of clinical metadata into multimodal learning pipelines.

Interpretability is ensured through Grad-CAM (Selvaraju et al., 2017), Integrated Gradients (Sundararajan et al., 2017), and anatomically guided retinal layer attribution.

Both frameworks are integrated into RETINAAI, a full-stack clinical decision support platform providing role-based access, AI inference pipelines, and explainability dashboards.

The principal contributions of this work are:

- A multimodal OCT diagnostic framework integrating retinal imaging and clinical metadata through dual cross-attention fusion, validated on the OCTDL benchmark.

- A metadata recovery module for handling missing, noisy, and inconsistent patient information in multimodal clinical AI systems.
- A hierarchical [GA-MIL](#)-based pipeline for weakly supervised patient-level [OCT](#) classification in real-world clinical settings.
- Integration of Grad-CAM, Integrated Gradients, and anatomically guided retinal layer attribution for interpretable AI predictions.
- RETINAAI, a deployable clinical decision support platform supporting secure workflows and AI-assisted diagnosis.
- Validation on both public benchmark and real-world clinical datasets, demonstrating robustness and generalization capability.

Dissertation Structure

This dissertation is organized into five chapters. **Chapter 1** surveys the relevant medical and technical literature, covering retinal pathologies, OCT imaging, deep learning, transformer architectures, multimodal AI, Multiple Instance Learning, and Explainable Artificial Intelligence (XAI). methods in ophthalmology. **Chapter 2** details the methodology, including the OCTDL multimodal framework, the clinical [Gated Attention \(GA\)-MIL](#) pipeline, preprocessing and recovery strategies, fusion mechanisms, and explainability approaches. **Chapter 3** presents the experimental results, encompassing quantitative evaluations, ablation studies, calibration analyses, and anatomical attribution experiments on both datasets. **Chapter 4** describes the RETINAI platform, covering software architecture, database design, role-based access control, AI integration, and clinical deployment workflows. **Chapter 5** evaluates the economic and strategic feasibility of RETINAI, including market analysis, Lean Canvas modeling, financial projections, and risk assessment. The dissertation concludes with a synthesis of findings, limitations, and future research directions.

Chapter 1

Background and State of the Art

This chapter presents the theoretical and clinical background necessary to understand the development of artificial intelligence systems for retinal disease analysis using optical coherence tomography (OCT). It introduces the fundamentals of retinal anatomy and major retinal diseases, reviews the principles and clinical applications of OCT imaging, and discusses the evolution of automated analysis methods from traditional image processing techniques to modern deep learning and transformer-based approaches.

1.1 Clinical Site Presentation: Ibn Al-Haytham Ophthalmology Center

The internship was carried out in collaboration with the Ibn Al-Haytham Ophthalmology Center, located in El Biar, Algiers. This structure is a specialized ophthalmology clinic dedicated to the diagnosis and follow-up of ocular diseases, including retinal pathologies.

The center provides advanced ophthalmic imaging and diagnostic services, including optical coherence tomography (OCT), and is equipped with modern medical imaging systems used in routine clinical practice.

This clinical environment provided valuable exposure to real-world ophthalmic workflows and contributed to a better understanding of practical challenges in retinal disease diagnosis and management.



Figure 1.1: Logo of the Ibn Al-Haytham Ophthalmology Center.

1.2 Retinal Diseases and Optical Coherence Tomography

The retina is a multilayered neural tissue lining the inner surface of the eye that converts light into electrical signals processed by the brain. It contains specialized layers including photoreceptors, bipolar cells, and ganglion cells. The macula is the central retinal region responsible for high-acuity vision, with the fovea at its center containing a high density of cone photoreceptors and no blood vessels, enabling sharp central vision.

Due to its functional importance, the macula is highly susceptible to diseases such as diabetic macular edema and age-related macular degeneration. [OCT](#) provides detailed visualization of retinal layers and is a key modality for assessing macular structure. The retinal anatomy is illustrated in [Figure 1.2](#).

Retinal diseases are a major cause of visual impairment and blindness worldwide, with prevalence increasing due to aging and systemic conditions such as diabetes. Early detection is essential to prevent irreversible vision loss ([Fujimoto and Swanson, 2016](#); [Abdi et al., 2025](#)).

[AMD](#) is a leading cause of central vision loss in older adults, affecting the macula. It is classified into dry and wet forms: dry [AMD](#) involves drusen accumulation and [Retinal Pigment Epithelium \(RPE\)](#) degeneration, while wet [AMD](#) is characterized by [Choroidal](#)

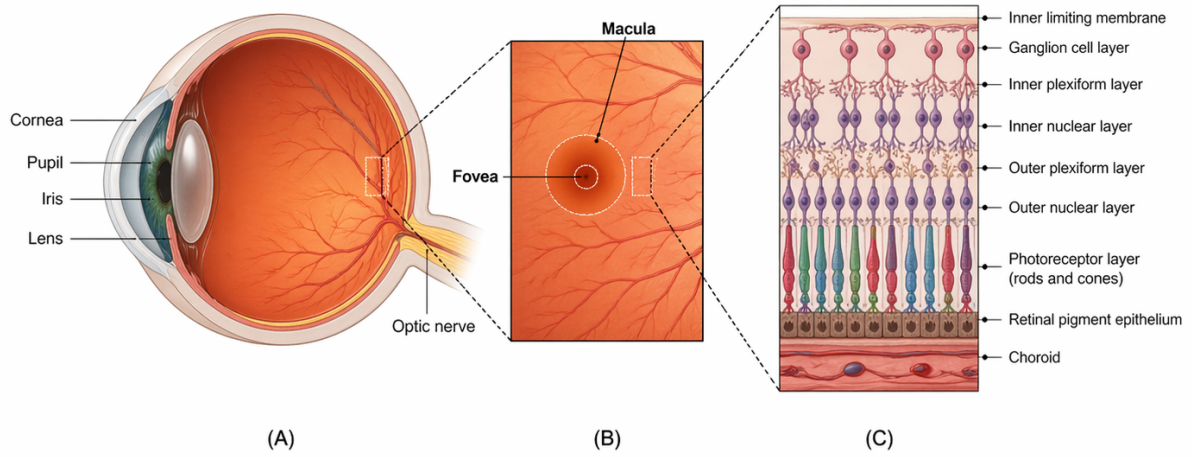


Figure 1.2: Anatomy of the human eye and retinal structure. The figure illustrates (A) a cross-sectional view of the eye, (B) the retinal fundus highlighting the macula and fovea, and (C) a layered representation of the retina.

Neovascularization (CNV) and fluid leakage. OCT is widely used to assess these structural changes and monitor progression (Kermany et al., 2018; He et al., 2025).

Diabetic Retinopathy (DR) and DME are major causes of blindness in working-age adults due to diabetes-induced microvascular damage leading to retinal fluid accumulation. OCT enables visualization of retinal thickening and cystoid spaces in DME, while early stages are often asymptomatic, motivating AI-assisted screening (Abràmoff et al., 2018; Fujimoto and Swanson, 2016).

OCT is a non-invasive imaging modality that provides cross-sectional visualization of retinal microstructures and enables detection of layer-level abnormalities not visible in fundus imaging (Huang et al., 1991a; Drexler and Fujimoto, 2008; Fujimoto and Swanson, 2016).

It allows identification of key biomarkers such as fluid accumulation, drusen, and retinal thickening in diseases including AMD and DME. These features form the basis of automated and deep learning-based diagnostic systems (Wang et al., 2024).

Clinically, OCT is widely used for diagnosing and monitoring retinal diseases by visualizing structural changes across retinal layers. In AMD, it detects drusen and fluid; in DME, retinal thickening and cystoid spaces; and in glaucoma, retinal nerve fiber layer and ganglion cell thickness (Kermany et al., 2018; Li et al., 2019).

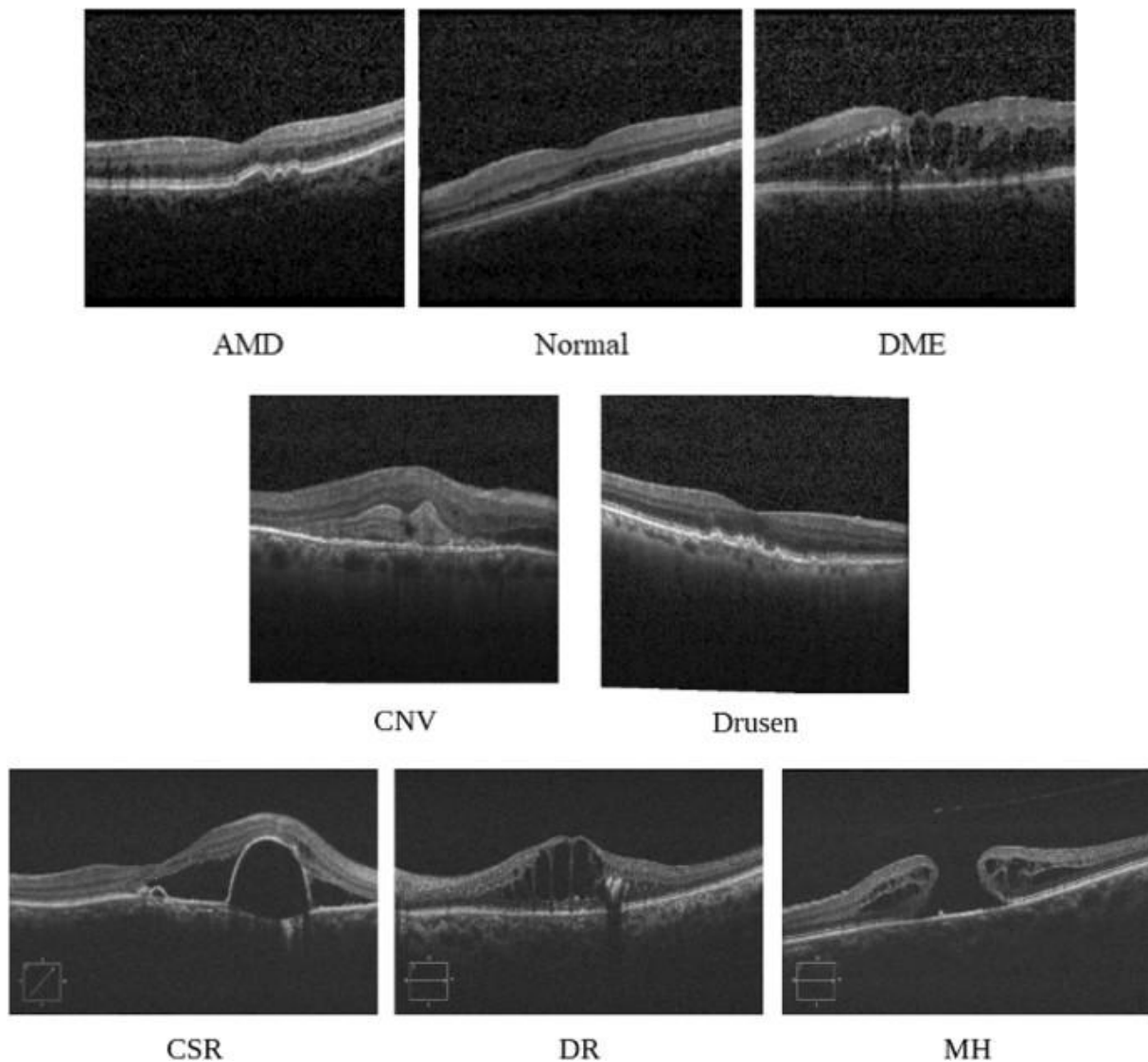


Figure 1.3: Representative OCT B-scan images showing normal retina and macular pathologies

Beyond diagnosis, it supports longitudinal monitoring and treatment evaluation by quantifying structural changes over time (De Fauw et al., 2018a; Abdi et al., 2025). It is also essential for guiding therapies such as anti-VEGF injections in neovascular AMD and DME, and is increasingly integrated into AI-based decision support systems (Abràmoff et al., 2018; Wang et al., 2024).

Despite its clinical importance, OCT interpretation remains challenging due to noise, subtle structural variations, and overlapping pathological patterns (Fujimoto and Swanson, 2016; Drexler and Fujimoto, 2008).

Inter-observer variability is common, particularly in complex diseases such as AMD

and DME, where interpretation depends on clinical experience (Bogunović et al., 2019; Venhuizen et al., 2020). In addition, volumetric OCT datasets are large, making manual analysis time-consuming and increasing the risk of missed findings (De Fauw et al., 2018a; Abdi et al., 2025).

Similar visual patterns across diseases further complicate differential diagnosis (Wang et al., 2024; He et al., 2025). These limitations motivate automated AI-based OCT analysis systems to improve consistency, scalability, and efficiency.

1.3 Segmentation in OCT Imaging

Retinal layer segmentation is a fundamental step in OCT analysis, enabling precise delineation of retinal structures and extraction of quantitative biomarkers for disease diagnosis and monitoring. Anatomical layers such as Retinal Nerve Fiber Layer (RNFL), Ganglion Cell-Inner Plexiform Layer (GCIPL), and outer retinal layers are strongly associated with major ophthalmic diseases including glaucoma, AMD, and DME (Li et al., 2019; Venhuizen et al., 2020).

Early segmentation methods relied on classical image processing techniques such as edge detection, graph-based optimization, and deformable models. While these approaches incorporated anatomical priors, they were highly sensitive to noise, low contrast, and pathological distortions, limiting their robustness in clinical settings (Drexler and Fujimoto, 2008; Fujimoto and Swanson, 2016).

The introduction of deep learning significantly improved segmentation performance. CNN-based encoder-decoder architectures, particularly U-Net, became the dominant paradigm due to their ability to combine high-level semantic information with precise spatial localization through skip connections (Ronneberger et al., 2015; Fang et al., 2017). Subsequent improvements such as CE-Net introduced multi-scale feature extraction and contextual encoding to better handle anatomical variability and disease-related deformations (Gu et al., 2019; Venhuizen et al., 2020).

More recently, transformer-based and hybrid CNN-transformer architectures have further advanced the field by modeling long-range dependencies and global context, leading

to improved performance in complex retinal structures and volumetric OCT data (Philippi et al., 2023; Hossain et al., 2023; Lu et al., 2024). As a result, segmentation has evolved into a robust and reliable component of automated OCT analysis pipelines.

Segmentation performance is typically evaluated using Dice coefficient and Intersection-over-Union (IoU), with recent methods achieving Dice scores exceeding 0.90 on benchmark datasets such as RETOUCH (Bogunović et al., 2019). Overall, these advances have made automated retinal segmentation a critical component of modern OCT-based clinical decision support systems (Abdi et al., 2025; Wang et al., 2024).

1.4 Automated Diagnosis of Retinal Diseases

1.4.1 Traditional Image Processing Methods

Before the advent of machine learning, OCT analysis relied on handcrafted image processing techniques to model retinal structures and extract diagnostic information. These methods used intensity gradients, texture patterns, and geometric constraints to identify anatomical and pathological features, but their performance was limited by noise, anatomical variability, and disease-induced distortions (Drexler and Fujimoto, 2008; Fujimoto and Swanson, 2016).

Early work focused on retinal layer segmentation using edge-based and optimization techniques. Methods such as Sobel and Canny filters were combined with graph-based optimization and dynamic programming to enforce anatomical consistency across B-scans (Lu et al., 2018). However, these approaches struggled in cases where severe pathology disrupted normal retinal boundaries.

For disease classification, handcrafted features such as retinal thickness, reflectivity profiles, and texture descriptors (e.g., LBP, HOG, Gabor filters, wavelets) were extracted and used with classical classifiers including support vector machines (SVMs), random forests, and k-nearest neighbors (Fujimoto and Swanson, 2016). SVMs were particularly effective in high-dimensional settings and achieved competitive performance in early OCT classification tasks.

Lesion detection methods typically relied on thresholding and region-growing techniques to identify fluid-filled regions. While effective for large and well-defined abnormalities, these methods were highly sensitive to noise and acquisition variability (Drexler and Fujimoto, 2008).

Overall, traditional approaches introduced early quantitative biomarkers but relied heavily on handcrafted rules, limiting their robustness and generalization. These limitations motivated the transition toward learning-based methods (Abdi et al., 2025; Wang et al., 2024).

1.4.2 Machine Learning Approaches

Machine learning methods improved OCT analysis by introducing data-driven classification frameworks while still relying on handcrafted feature extraction. A typical pipeline included preprocessing, segmentation, feature engineering, and supervised classification (Abdi et al., 2025; Wang et al., 2024).

Features such as retinal thickness, reflectivity profiles, and texture descriptors (e.g., LBP, HOG, wavelets) were widely used with classifiers such as SVMs, random forests, k-nearest neighbors, and logistic regression. Among these, SVMs remained popular due to their robustness in small datasets and high-dimensional feature spaces.

These methods were also applied to clinically relevant biomarkers, such as RNFL and GCIPL thickness, particularly in glaucoma detection (Li et al., 2019). However, performance was strongly dependent on feature quality and segmentation accuracy, making them sensitive to noise and inter-device variability.

With the availability of large OCT datasets, machine learning approaches were progressively replaced by deep learning methods, which learn representations directly from images and achieve superior generalization and accuracy (Kermany et al., 2018; De Fauw et al., 2018a; Abdi et al., 2025).

1.4.3 Deep Learning Approaches for OCT Classification

Deep learning has enabled end-to-end OCT classification by learning hierarchical representations directly from raw images, eliminating the need for handcrafted features. CNNs are the dominant architecture due to their ability to capture both local structures and global pathology patterns (Kermany et al., 2018; Abdi et al., 2025).

A landmark study by Kermany et al. (2018) demonstrated the effectiveness of an Inception-v3 CNN for classifying OCT scans into normal, drusen, DME, and CNV categories, achieving over 96% accuracy and establishing the feasibility of automated screening (Kermany et al., 2018).

Subsequent studies explored architectures such as VGG, ResNet, DenseNet, and Inception, where deeper residual networks improved feature representation and robustness across multiple retinal diseases (He et al., 2016; Simonyan and Zisserman, 2015; Li et al., 2019).

To exploit volumetric OCT data, 3D CNNs were introduced to model inter-slice dependencies, improving performance in glaucoma and macular disease classification tasks (Maetschke et al., 2019; Kumar et al., 2021).

More recently, attention mechanisms and hybrid CNN–transformer models have been proposed to enhance global context modeling and improve interpretability, achieving state-of-the-art performance on large-scale datasets (He et al., 2023; Oghbaie et al., 2024; Wang et al., 2024).

Overall, OCT classification has evolved from 2D CNNs to volumetric and attention-based architectures, enabling scalable and accurate automated diagnosis systems (Abdi et al., 2025; Devine et al., 2025).

1.4.4 Transformer-based Approaches for OCT Analysis

Transformer-based models have recently emerged as powerful alternatives to CNNs for OCT analysis due to their ability to model long-range dependencies via self-attention. Unlike convolutional models that focus on local receptive fields, transformers process images as sequences of patches, enabling global context modeling across retinal structures

(Dosovitskiy et al., 2021; Abdi et al., 2025).

Early adaptations of ViT demonstrated competitive performance in OCT segmentation and classification tasks. For instance, ViT-based models have been applied to retinal lesion segmentation in neovascular AMD, achieving results comparable to CNNs while using fewer inductive biases (Philippi et al., 2023).

To improve computational efficiency, hybrid architectures such as the Swin Transformer introduced shifted-window attention, reducing complexity while maintaining strong contextual modeling. Extensions such as Swin-Poly combine CNN-based local feature extraction with transformer-based global reasoning, improving robustness in OCT analysis (He et al., 2023; Liu et al., 2021).

Transformer architectures have also been extended to volumetric OCT data. Swin-VFTR uses 3D attention mechanisms to model inter-slice relationships for fluid segmentation, while full-attention models such as FAT enable end-to-end OCT volume classification by jointly capturing local and global representations (Hossain et al., 2023; Oghbaie et al., 2024).

Overall, transformer-based approaches improve global context modeling and interpretability compared to CNNs. Although computational cost and generalization across devices remain challenges, hybrid and efficient attention-based models are increasingly shaping the future of OCT analysis systems (Wang et al., 2024; Devine et al., 2025).

1.5 Multimodal Learning in Medical Imaging

1.5.1 Integration of Clinical Data and Imaging Data

Integrating clinical data with medical imaging improves diagnostic accuracy by incorporating complementary patient information such as demographics, systemic conditions, disease history, and visual acuity. In ophthalmology, clinicians naturally combine OCT interpretation with such contextual information, motivating multimodal learning approaches that jointly model imaging and structured clinical metadata (Wang et al., 2024; Abdi et al., 2025).

Early multimodal approaches typically relied on late fusion, where features extracted from OCT images using CNNs were combined with clinical variables at the decision stage. While simple and effective, this strategy limits interaction between modalities and may fail to capture relationships between imaging patterns and patient-specific factors (Morano et al., 2024).

To address this limitation, early and intermediate fusion strategies were introduced to enable deeper interaction between modalities during representation learning, improving robustness and diagnostic performance (Atik et al., 2024). More recently, attention-based and cross-modal learning methods have been proposed to dynamically integrate modalities by allowing clinical information to guide visual feature learning (Jin et al., 2021).

Projection-based methods further enhance multimodal integration by mapping imaging and clinical features into a shared latent space, enabling more consistent representations for disease prediction and prognosis modeling (Morano et al., 2024). Overall, multimodal learning provides a clinically aligned framework for OCT analysis by combining retinal imaging with patient context (Wang et al., 2024; Abdi et al., 2025).

1.5.2 Fusion Strategies for Multimodal Diagnosis

Multimodal fusion defines how imaging and clinical data are integrated within a learning model and plays a key role in determining performance and interpretability (Wang et al., 2024; Abdi et al., 2025).

Early fusion combines modalities at the input level before feature extraction. This allows joint representation learning but can be challenging due to differences in data structure, scale, and noise characteristics (Atik et al., 2024).

Late fusion processes each modality independently using dedicated encoders (e.g., CNNs for OCT and MLPs for clinical data) and combines predictions at the decision level. Although flexible and widely used, it limits cross-modal interaction and may overlook complementary relationships between modalities (Morano et al., 2024).

Intermediate (hybrid) fusion enables interaction between modalities at feature levels, improving cross-modal reasoning and representation quality. Attention-based fusion

further enhances this process by dynamically weighting features based on relevance. Cross-attention mechanisms, for example, have been successfully applied to combine OCT and OCT angiography for improved diagnosis (Jin et al., 2021).

Projection-based fusion methods extend this idea by learning a shared latent space for imaging and clinical features, encouraging aligned multimodal representations and improving predictive performance (Morano et al., 2024).

Overall, while late fusion remains a common baseline, intermediate and attention-based strategies provide stronger cross-modal interaction and are increasingly preferred in multimodal OCT analysis systems (Wang et al., 2024; Abdi et al., 2025).

1.5.3 Attention Mechanisms and Cross-Modal Interaction

Attention mechanisms improve deep learning models by enabling them to focus on the most relevant features within high-dimensional data. In OCT analysis, they enhance the representation of clinically important structures such as retinal layers, lesions, and fluid regions (Selvaraju et al., 2020; Oktay et al., 2019).

In unimodal settings, attention modules such as Attention U-Net improve segmentation performance by selectively emphasizing relevant anatomical regions while suppressing background noise (Oktay et al., 2019).

In multimodal learning, attention extends to cross-modal interaction, where information from different modalities is dynamically fused. This allows clinical metadata to influence visual feature extraction, aligning model behavior more closely with clinical reasoning (Wang et al., 2024; Abdi et al., 2025).

Recent retinal imaging studies have explored multimodal fusion using OCT and OCT angiography, demonstrating improved performance through richer representations of retinal pathology (Jin et al., 2021). Additional work integrates OCT with clinical or demographic data using feature-level or late fusion strategies to improve robustness.

Transformer-based architectures naturally support cross-modal interaction through self-attention and cross-attention mechanisms, enabling bidirectional information exchange and improved contextual understanding (Dosovitskiy et al., 2021; Liu et al., 2021).

Despite these advances, most current OCT multimodal systems still rely on simple concatenation or late fusion, while fully learned cross-attention between imaging and clinical data remains underexplored. Moreover, attention-based explanations should be interpreted carefully, as attention maps do not always directly correspond to clinically meaningful regions (Selvaraju et al., 2020; Abdi et al., 2025).

1.6 Explainable AI in Ophthalmology

1.6.1 Visual Explainability Techniques

Visual explainability methods are widely used in OCT-based deep learning systems to interpret predictions and improve clinical trust. These techniques identify image regions that contribute most to model decisions, ensuring that predictions are based on meaningful retinal structures rather than spurious correlations (Abdi et al., 2025; Wang et al., 2024).

Gradient-based saliency methods estimate the sensitivity of predictions with respect to input pixels. Although they provide a first level of interpretability, basic saliency maps are often noisy and lack precise localization (Selvaraju et al., 2020).

Grad-CAM is one of the most widely adopted methods in OCT analysis. It generates class-specific heatmaps by weighting feature maps using gradient information, highlighting clinically relevant structures such as intraretinal cysts, subretinal fluid, and pigment epithelial detachments (Selvaraju et al., 2020; Abdi et al., 2025).

Model-agnostic approaches such as LIME and SHAP provide complementary explanations by estimating feature importance independently of the underlying model. These methods are commonly used to validate whether predictions rely on anatomically meaningful regions (Ribeiro et al., 2016; Lundberg and Lee, 2017).

Attention-based visualization techniques offer another form of interpretability by highlighting spatial regions or feature channels that contribute most to predictions. In OCT analysis, attention maps often align with clinically relevant structures such as retinal layers and lesion areas (Oktay et al., 2019; Zhang et al., 2020).

Despite their usefulness, visual explanations should be interpreted with caution, as

highlighted regions do not necessarily imply causal relationships. Reliable interpretation typically requires complementary evaluation such as expert validation and quantitative analysis (Wang et al., 2024; Abdi et al., 2025).

1.6.2 Multimodal Interpretability Methods

With the integration of imaging and clinical data, interpretability has expanded toward explaining how multiple modalities jointly contribute to model predictions. Multimodal interpretability aims to quantify both modality-specific contributions and cross-modal interactions (Wang et al., 2024; Abdi et al., 2025).

Modality-level attribution methods estimate the relative contribution of each data source (e.g., OCT images vs clinical variables) to the final prediction. Extensions of SHAP and similar methods enable separate importance scoring for imaging and structured clinical data, helping identify how patient-specific factors influence model output (Lundberg and Lee, 2017; Morano et al., 2024; Atik et al., 2024).

Feature-level interpretability combines visual explanations such as Grad-CAM with tabular feature importance measures, enabling joint analysis of retinal structures and clinical variables influencing predictions (Selvaraju et al., 2020; Wang et al., 2024).

Attention-based multimodal models further enhance interpretability by revealing interactions between modalities. Cross-attention mechanisms, for instance, allow clinical metadata to guide focus on relevant OCT regions, providing insight into how patient context influences visual interpretation (Jin et al., 2021; Chen et al., 2022).

However, multimodal explanations can be complex and may not always reflect causal relationships. As a result, robust interpretation often requires complementary validation strategies such as ablation studies and clinical evaluation (Abdi et al., 2025; Wang et al., 2024).

1.7 Publicly Available OCT Datasets and Clinical Data

1.7.1 Overview of Public OCT Datasets

Public OCT datasets are essential for developing and evaluating machine learning models in ophthalmology, providing standardized benchmarks with annotated retinal images for segmentation, classification, and multimodal learning tasks.

One of the most widely used datasets is the UCSD OCT dataset introduced by Kermany et al. (2018), which contains over 100,000 labeled macular OCT B-scans across major disease categories such as CNV, DME, drusen, and normal cases (Kermany et al., 2018).

The OCTDL dataset extends this resource by providing both 2D and 3D OCT data from multiple imaging devices, along with clinical metadata and detailed annotations, enabling multimodal and multi-task learning research (Kulyabin et al., 2024).

Other important datasets include the Duke OCT dataset, which provides volumetric scans with expert retinal layer annotations for structural analysis, and the Stanford OCT dataset, which contains labeled macular scans for diseases such as AMD and DME (Maaß et al., 2015; Kermany et al., 2018).

Benchmark challenges such as RETOUCH provide multi-vendor OCT volumes annotated for retinal fluid segmentation, including intraretinal, subretinal, and pigment epithelial detachment regions. These datasets are widely used for evaluating segmentation algorithms under standardized conditions (Bogunović et al., 2019).

More recent MICCAI challenges have introduced multi-center OCT datasets to evaluate model generalization across devices and clinical settings, further improving robustness assessment in real-world scenarios (Abdi et al., 2025; Wang et al., 2024).

1.7.2 Data Preprocessing Strategies

Data preprocessing is a critical step in OCT-based analysis, as raw scans are often affected by speckle noise, acquisition variability, and anatomical distortions that can degrade downstream learning performance. Effective preprocessing improves image quality and

ensures standardized inputs for robust machine learning models (Drexler and Fujimoto, 2008; Fujimoto and Swanson, 2016).

Core preprocessing operations include denoising, retinal normalization, intensity standardization, registration, and quality assessment. Denoising techniques aim to reduce speckle noise using classical methods such as median filtering, wavelet transforms, and anisotropic diffusion, as well as more recent learning-based approaches that provide improved robustness on low-quality scans (Drexler and Fujimoto, 2008; Li et al., 2022).

Retinal normalization, including retinal flattening, reduces anatomical variability by aligning a reference layer—typically the retinal pigment epithelium (RPE)—thereby improving structural consistency across scans (Venhuizen et al., 2020). Intensity standardization mitigates variations introduced by different imaging devices using techniques such as histogram equalization or z-score normalization, which enhances cross-dataset generalization (Kermany et al., 2018).

In multimodal and longitudinal settings, image registration ensures spatial alignment across scans acquired at different time points or modalities. Quality assessment methods further improve dataset reliability by detecting and filtering motion artifacts and low-quality acquisitions prior to analysis (Fang et al., 2019; Devalla et al., 2018).

Image Preprocessing

Image preprocessing focuses on enhancing scan quality and standardizing spatial resolution for downstream learning tasks. Speckle noise reduction is typically performed using Gaussian, median, or wavelet-based filtering methods (Drexler and Fujimoto, 2008; Li et al., 2022).

Retinal flattening and intensity normalization further reduce anatomical and acquisition variability, improving the robustness of segmentation and classification models (Venhuizen et al., 2020; Kermany et al., 2018). Additional operations such as resizing and cropping are applied to ensure uniform input dimensions and to emphasize clinically relevant retinal regions.

Quality control procedures, including motion artifact detection and scan quality fil-

tering, are used to remove corrupted or unreliable images, thereby increasing dataset consistency and model stability (Fang et al., 2019; Devalla et al., 2018).

Clinical Data Preprocessing

Clinical metadata preprocessing is essential in multimodal OCT systems, where patient information is often heterogeneous, incomplete, and inconsistently recorded. Typical variables include demographic attributes, systemic conditions, clinical history, and treatment information (Morano et al., 2024; Atik et al., 2024).

Missing values are handled using standard imputation strategies such as mean/median substitution or k-nearest neighbors imputation, depending on feature type (Wang et al., 2024). Numerical variables are standardized using normalization techniques such as z-score scaling, while categorical variables are encoded using one-hot representations or learned embeddings in deep learning frameworks (Holland and Smith, 2022).

Additional preprocessing steps, including outlier removal, data cleaning, and temporal alignment, ensure consistency between clinical records and corresponding OCT acquisitions, particularly in longitudinal studies (Morano et al., 2024).

1.8 Key Insights and Research Gaps

Deep learning has significantly advanced OCT-based retinal analysis, enabling accurate automated segmentation, classification, and structural assessment. CNN-based methods established strong baselines by learning directly from OCT images and achieving high diagnostic performance across diseases such as AMD, DME, and glaucoma (Kermany et al., 2018; De Fauw et al., 2018a; Li et al., 2019). Subsequent works introduced multi-scale, multi-task, and 3D architectures that better capture local retinal structures and global spatial context (Venhuizen et al., 2020; Maetschke et al., 2019).

More recently, transformer-based models have gained attention due to their ability to model long-range dependencies and improve global context understanding. Vision transformers and hybrid CNN–transformer architectures have demonstrated strong performance in segmentation and volumetric OCT analysis (Philippi et al., 2023; He et al.,

2023; Lu et al., 2024; Hossain et al., 2023). In parallel, explainability methods such as saliency maps and attention visualization are increasingly used to improve transparency in clinical AI systems (Selvaraju et al., 2020; Wang et al., 2024).

Despite these advances, key gaps remain. Most existing OCT models are image-centric and do not fully integrate clinical information such as demographics or patient history. Current multimodal approaches often rely on simple fusion strategies that fail to capture complex imaging-clinical interactions (Morano et al., 2024; Atik et al., 2024; Jin et al., 2021; Chen et al., 2022).

Another limitation is poor cross-device generalization, as models trained on specific datasets often degrade on OCT scans from different scanners or clinical centers (Bogunović et al., 2019; Li et al., 2020; Wang et al., 2024). Although attention-based explanations improve interpretability, their clinical validity is not always rigorously verified (Selvaraju et al., 2020; Oktay et al., 2019).

Finally, relatively few studies address longitudinal OCT analysis and disease progression modeling, which are essential for treatment planning and long-term monitoring (Abdi et al., 2025; Devine et al., 2025).

Overall, these limitations highlight the need for clinically aligned, multimodal, and generalizable OCT-based frameworks that integrate imaging and clinical data while ensuring robustness and reliable interpretability (Abdi et al., 2025; Wang et al., 2024).

1.9 Conclusion

This chapter provided a comprehensive overview of retinal anatomy, optical coherence tomography (OCT), and the evolution of automated approaches for retinal disease analysis. It first introduced the clinical background of major retinal pathologies such as age-related macular degeneration (AMD), diabetic macular edema (DME), and diabetic retinopathy, highlighting the critical role of OCT imaging in their diagnosis, monitoring, and management.

The discussion then addressed key preprocessing strategies required to handle the inherent noise, variability, and heterogeneity of real-world OCT data. Classical and

deep learning-based approaches for segmentation and classification were subsequently reviewed, illustrating the progression from handcrafted feature engineering to advanced CNN, transformer, and hybrid architectures capable of learning robust hierarchical representations.

In addition, the chapter examined multimodal learning strategies that integrate imaging data with clinical metadata, as well as explainable artificial intelligence techniques that enhance interpretability and clinical trust. Publicly available OCT datasets were also presented, along with their role in benchmarking and advancing automated retinal analysis systems.

Despite recent advances, challenges remain in achieving robust multimodal integration, ensuring generalization across different imaging devices and clinical settings, and improving the reliability of model interpretability in real-world applications. These limitations motivate the development of clinically aligned, context-aware frameworks that jointly model imaging and patient-level data while ensuring robustness, interpretability, and practical applicability in ophthalmic workflows.

Chapter 2

Methodology

2.1 Introduction

This chapter presents the methodological frameworks developed for automated macular disease diagnosis using OCT images and structured clinical metadata. The proposed methodology is organized into two complementary contributions, each designed for a distinct objective and data setting.

The first contribution is a research-oriented experimental framework built on a publicly available OCT benchmark dataset. It is designed to study multimodal representation learning under controlled experimental conditions, enabling systematic evaluation of different learning strategies and model configurations.

The second contribution is a clinically oriented decision support framework developed using a proprietary dataset collected at the Ibn Al Haythem Ophthalmology Center. It is designed to operate under real-world clinical constraints and focuses on robust and reliable disease prediction in practical settings.

Across both contributions, the study investigates image-only, multimodal, and patient-level learning paradigms, ensuring a consistent comparative evaluation across different levels of information integration.

2.2 Problem Formulation

This work addresses automated macular disease classification from OCT data through three complementary learning paradigms: image-level classification, multimodal learning with structured clinical metadata, and patient-level MIL. Each formulation corresponds to a different level of data granularity and supervision, and is evaluated independently to analyze the impact of modality integration and annotation structure on diagnostic performance.

It is important to note that these paradigms are not uniformly applied across all frameworks. In particular, MIL is exclusively adopted in Framework II (Clinical Decision Support Strategy), which is based on a real-world dataset, where supervision is limited to patient-level labels and slice-level annotations are unavailable. In contrast, Framework I is built on the OCTDL benchmark dataset, which provides slice-level annotations, making MIL unnecessary in that setting.

2.2.1 Image-Level Classification

In the image-level setting, each OCT B-scan is represented as $x \in \mathbb{R}^{H \times W}$, with an associated disease label $y \in \mathcal{C}$, where $\mathcal{C}$ denotes the set of target retinal disease categories. The objective is to learn a discriminative mapping:

$$f_{\theta} : x \rightarrow y$$

This formulation assumes fully supervised learning at the slice level and serves as a baseline for subsequent experiments.

2.2.2 Multimodal Learning

In the multimodal setting, each sample is represented as a pair (x, m) , where x denotes the OCT image and $m \in \mathbb{R}^d$ is a structured clinical metadata vector including attributes

such as age, sex, and eye laterality. The learning objective is defined as:

$$f_{\theta} : (x, m) \rightarrow y$$

The model learns a joint representation by encoding each modality separately and integrating them through a fusion mechanism. This allows the network to exploit complementary information from both imaging data and clinical context.

2.2.3 Patient-Level Multiple Instance Learning

In the patient-level setting, each OCT examination is represented as a bag of B-scan slices:

$$\mathcal{V}_i = \{x_{i1}, x_{i2}, \dots, x_{iT_i}\}$$

where T_i denotes the number of slices in volume i . Supervision is available only at the patient (volume) level, and slice-level annotations are not provided.

The learning objective is:

$$f_{\theta} : \mathcal{V}_i \rightarrow y_i$$

To handle this weakly supervised setting, a MIL framework is adopted, where slice-level embeddings are aggregated into a patient-level representation using an attention-based pooling mechanism. This enables the model to learn from incomplete supervision while identifying diagnostically informative slices within each volume.

2.3 Dataset Strategy

This study adopts a dual-dataset strategy aligned with the two methodological contributions. A publicly available OCT benchmark dataset is used for controlled experimental evaluation under the Framework I, while a real-world clinical OCT dataset supports patient-level learning and clinical validation under the Framework II.

2.3.1 Public OCT Benchmark Dataset

The Optical Coherence Tomography Dataset for Image-Based Deep Learning Methods (OCTDL) (Kulyabin et al., 2023) is a publicly available benchmark designed for automated retinal disease classification. The dataset comprises over 2,000 annotated OCT B-scan images spanning multiple macular and retinal conditions: AMD, DME, Epiretinal Membrane (ERM), Retinal Vein Occlusion (RVO), Retinal Artery Occlusion (RAO), Vitreoretinal Interface Disorder (VID), and Normal (NO) cases.

The dataset exhibits significant class imbalance, with AMD constituting the majority class while conditions such as RAO and VID are substantially underrepresented. This characteristic renders OCTDL suitable for evaluating model behavior under imbalanced learning conditions. Data are provided in two complementary formats: OCT images and a structured metadata file containing diagnostic labels and associated acquisition attributes. In this study, each B-scan is treated as an independent sample for image-level and multimodal classification experiments.

2.3.2 Real-World Clinical OCT Dataset

The clinical dataset was assembled in collaboration with the Ibn Al Haythem Center and consists of anonymized OCT volumes acquired during routine ophthalmological examinations. Each patient is represented by a set of B-scan slices extracted from macular volumes, with varying acquisition completeness depending on clinical protocols and imaging conditions.

The dataset supports two levels of supervision. A subset carries patient-level binary labels for normal/abnormal screening, while a smaller subset includes expert-validated fine-grained annotations for disease-specific classification. Labels are assigned at the patient or volume level, and no individual slice annotations are available, necessitating a weakly supervised MIL approach.

The dataset reflects the full complexity of real-world clinical data, including variability in acquisition protocols, image quality heterogeneity, and demographic diversity. Clinical metadata attributes, including patient age and sex, are only partially available depend-

ing on the completeness of individual medical records. This dataset is used exclusively within Framework II to evaluate patient-level diagnostic learning under realistic clinical constraints.

2.4 Evaluation Protocol

The evaluation protocol is designed to provide a comprehensive and clinically meaningful assessment of model performance. Given the medical nature of the classification task and the presence of class imbalance, accuracy alone is not used as the primary evaluation criterion, in favor of more discriminative and robust metrics.

2.4.1 Performance Metrics

Model performance is assessed using Precision, Recall, F1-score, [ROC-AUC](#), and Confusion Matrix analysis. [ROC-AUC](#) is computed using a one-vs-rest strategy in multi-class settings, with class-wise scores macro-averaged. These metrics collectively enable assessment of both overall classification performance and class-wise error behavior under imbalanced conditions.

2.4.2 Calibration Analysis

Predictive calibration is evaluated using reliability diagrams and [Expected Calibration Error \(ECE\)](#), ensuring that model confidence scores are well aligned with empirical outcome frequencies. Calibration assessment is particularly important in clinical decision support, where poorly calibrated probability estimates may lead to inappropriate risk stratification.

2.4.3 Patient-Level Evaluation in the MIL Setting

For the clinical dataset, evaluation is performed at the patient level by aggregating slice-level predictions across each [OCT](#) volume prior to metric computation. All performance

metrics are reported against patient-level ground truth labels to accurately reflect real clinical decision-making.

2.4.4 Cross-Framework Evaluation Consistency

To ensure fair and comparable evaluation across experiments, all models are assessed using identical data splits and preprocessing pipelines. Hyperparameter selection and model checkpoint selection are performed exclusively on the validation set, while final performance is reported on a held-out test set to prevent data leakage and ensure unbiased evaluation.

2.5 Explainability and Interpretability

To improve the transparency and clinical interpretability of the multimodal diagnostic framework, a dedicated explainability module is incorporated into the inference pipeline. Since clinical adoption of deep learning systems requires not only strong predictive performance but also interpretable decision-making, multiple complementary explainability techniques are employed to analyze both retinal imaging biomarkers and structured clinical feature contributions.

2.5.1 Visual Explainability via Grad-CAM

[Grad-CAM](#) is applied to the image branch of the multimodal framework to identify the retinal regions contributing most significantly to model predictions. Since the proposed architecture employs a Swin Transformer as the visual encoder, the [Grad-CAM](#) implementation is adapted to transformer-based feature representations. Feature activations and gradient signals are extracted from intermediate transformer layers using forward and backward hooks, and the resulting activation maps are projected onto the original [OCT](#) images to produce spatially coherent heatmaps highlighting discriminative retinal structures.

2.5.2 Multimodal Attribution via Integrated Gradients

Integrated Gradient is applied to the full multimodal architecture to quantify the contribution of both input modalities to the final prediction. Attribution scores are computed independently for the OCT image branch and the clinical metadata branch. For the visual branch, pixel-level attribution maps identify diagnostically relevant retinal structures. For the metadata branch, feature-level attribution scores are assigned to each clinical variable (age, sex, and eye laterality), enabling quantitative assessment of their respective contributions to the diagnostic output.

2.5.3 Anatomically Guided Retinal Layer Attribution

To provide clinically grounded spatial interpretation, Grad-CAM activation maps are further combined with retinal layer segmentation using a DeepLabV3 model with a ResNet-50 backbone. The segmentation framework delineates seven major retinal anatomical layers: Internal Limiting Membrane (ILM), RNFL, Inner Plexiform Layer (IPL), Inner Nuclear Layer (INL), Outer Plexiform Layer (OPL), Outer Nuclear Layer (ONL), and RPE. The generated segmentation masks are spatially aligned with Grad-CAM activation maps, enabling layer-wise attribution analysis by computing mean attention intensity within each anatomical region. This anatomically guided framework provides a clinically interpretable decomposition of model decisions at the structural layer level.

Together, Grad-CAM, Integrated Gradients, and anatomically guided layer attribution constitute a layered explainability pipeline that provides spatial, multimodal, and anatomically grounded interpretation of model predictions, thereby improving clinical reliability and transparency.

2.6 Preprocessing

2.6.1 Shared Data Organization and Partitioning Strategy

A shared data organization and partitioning strategy is defined across both frameworks to ensure reproducibility and fair evaluation. All OCT samples are structured into a machine-learning-ready format, associating each image with its corresponding diagnostic label and available metadata. Patient identifiers are preserved solely for grouping purposes in patient-level experiments and are explicitly excluded from model inputs.

A patient-wise splitting strategy is applied to both datasets to prevent data leakage across subsets. All samples belonging to a given patient are assigned exclusively to a single partition, ensuring that the model is evaluated on entirely unseen patients. The resulting splits follow an 80%, 10%, 10% training, validation, test ratio.

Training set class imbalance is addressed using random oversampling with replacement. No synthetic data generation techniques are employed in order to preserve the authenticity of the medical data distribution. To further improve generalization, on-the-fly data augmentation is applied exclusively during training, including random horizontal flipping, small-angle rotations ($\pm 10^\circ$), and stochastic variations in brightness and contrast. Augmentation is disabled during validation and testing to ensure evaluation integrity.

Random oversampling is preferred over synthetic data generation methods such as SMOTE to avoid introducing artificially generated samples that may distort the underlying medical image distribution. It is also favored over cost-sensitive loss weighting alone, which, without explicit rebalancing of the training distribution, may still lead to suboptimal representation learning for minority classes in mini-batch optimization.

Although oversampling increases the risk of overfitting to minority-class samples, this effect is mitigated through on-the-fly data augmentation, which introduces stochastic variability to oversampled instances across training epochs, thereby improving robustness and generalization.

2.6.2 Framework I: Image and Metadata Preprocessing

Image Preprocessing in the Benchmark Dataset

Raw OCT B-scans from the OCTDL dataset exhibit acquisition-related inconsistencies including speckle noise and inter-scan intensity variability. To standardize inputs across experiments, all images are resized to 224×224 pixels to ensure compatibility with the selected deep learning architectures. Median filtering is applied to suppress speckle noise while preserving the structural boundaries of retinal layers. Cropping is intentionally avoided, as both central and peripheral retinal regions carry diagnostically relevant information. Pixel intensities are subsequently normalized to the $[0, 1]$ range prior to model input.

Clinical Metadata Recovery in the Benchmark Dataset

The metadata associated with the OCTDL dataset exhibits significant semantic inconsistencies. Specifically, the age attribute was incorrectly encoded in 1,616 out of 2,064 samples (78.3%), with valid chronological ages replaced by a constant artifact value (2023). Sex and eye laterality fields were similarly affected by inconsistent encoding schemes, yielding ambiguous or corrupted categorical entries.

To address these inconsistencies, a dedicated metadata Recovery module is introduced to recover corrupted or missing attributes directly from the corresponding OCT B-scan images. This module is trained independently from the downstream multimodal classification pipeline and is not invoked during inference. A task-specific model is assigned to each attribute: a Swin Transformer-B for age regression, a hybrid DenseNet-121 and Transformer architecture for sex classification, and a second Swin Transformer-B for eye laterality classification.

Task-specific models were preferred over a shared multi-task architecture due to the divergent nature of the three prediction objectives — regression for age, and binary classification for sex and laterality — which impose different loss scales, label granularities, and convergence dynamics that are difficult to reconcile within a single joint training objective.

It is important to emphasize that the reconstructed values are not treated as ground-truth clinical measurements. They serve as auxiliary contextual variables intended to reduce semantic inconsistencies in the dataset, and their role remains supportive rather than diagnostic. The downstream classification framework is primarily driven by OCT imaging features.

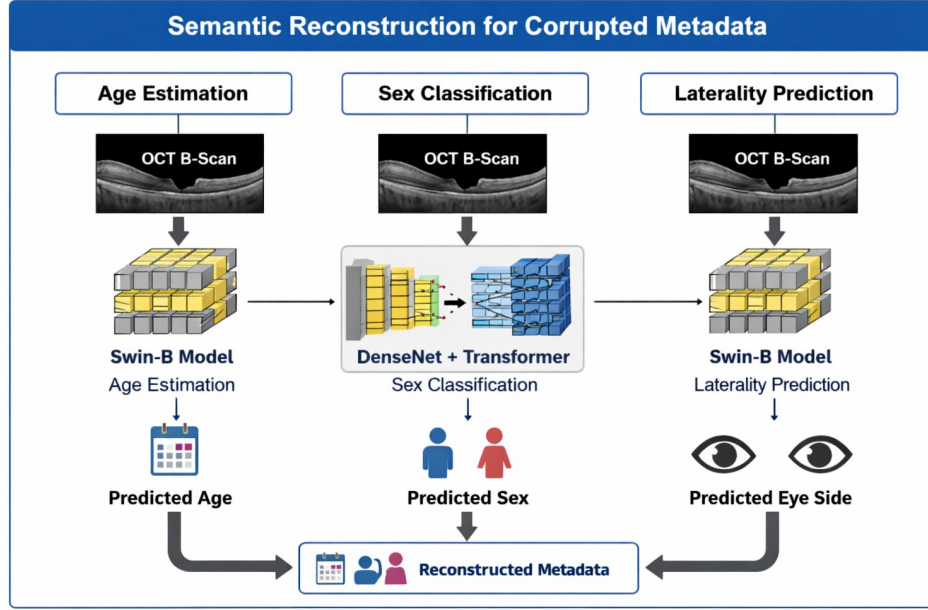


Figure 2.1: Overview of the clinical metadata Recovery module used to recover corrupted attributes in the OCTDL dataset.

The reconstructed attributes are combined with valid metadata entries to form a consistent dataset of 2,064 samples. To prevent information leakage, Recovery is performed independently within each dataset split, maintaining strict train/validation/test isolation.

2.6.3 Framework II: Image and Metadata Preprocessing

Image Preprocessing in the Real-World Dataset

A dedicated preprocessing pipeline is applied to transform raw clinical OCT scans into a standardized representation suitable for deep learning-based analysis. The pipeline begins by filtering out corrupted or non-OCT images based on decoding errors and metadata inconsistencies. Contrast Limited Adaptive Histogram Equalization (CLAHE) is then applied to enhance local retinal contrast and improve the visibility of pathological structural

changes.

A [Region of Interest \(ROI\)](#) extraction step is subsequently employed to suppress irrelevant background regions while retaining anatomically meaningful retinal structures, thereby directing model attention toward diagnostically relevant areas of the scan. All images are resized to 384×384 pixels to ensure uniform input dimensions, and pixel intensities are normalized to the $[0, 1]$ range prior to model input. Each processed sample is stored alongside its associated metadata and patient identifier to ensure full traceability and reproducibility.

Clinical Metadata Preprocessing in the Real-World Dataset

Clinical metadata for Framework II is constructed using expert-validated annotations provided by ophthalmologists, ensuring the reliability of the supervision signal. Image-metadata alignment is performed at the patient level through identifier mapping, with consistency checks applied to avoid cross-patient mismatches.

Standard data cleaning procedures are performed, including deduplication, resolution of conflicting records, and exclusion of incomplete entries. Missing demographic attributes are handled according to feature type: patient age is derived from birth date records when available and standardized using z-score normalization, while categorical variables such as sex are numerically encoded with an explicit *unknown* category to handle missing values without introducing systematic bias.

All preprocessing transformations, including encoders and normalization parameters, are stored to ensure reproducibility during both training and inference. The final dataset comprises [OCT](#) image paths, patient identifiers, structured clinical metadata, and expert-confirmed diagnostic labels, fully prepared for patient-level [MIL](#) experiments.

2.7 Framework I: Multimodal Learning Strategies

Framework I is a research-oriented experimental framework designed to systematically evaluate multimodal representation learning strategies for [OCT](#)-based macular disease classification. The primary objective is to isolate and quantify the contribution of in-

dividual architectural components through a structured ablation study conducted under controlled experimental conditions using the OCTDL benchmark dataset.

The proposed architecture is decomposed into three core modules: image encoding, metadata encoding, and multimodal fusion. These are evaluated separately and in combination, with all experiments modifying a single component at a time while keeping the remaining modules fixed. Figure 2.2 provides an overview of the full framework and illustrates the components evaluated during the ablation study.

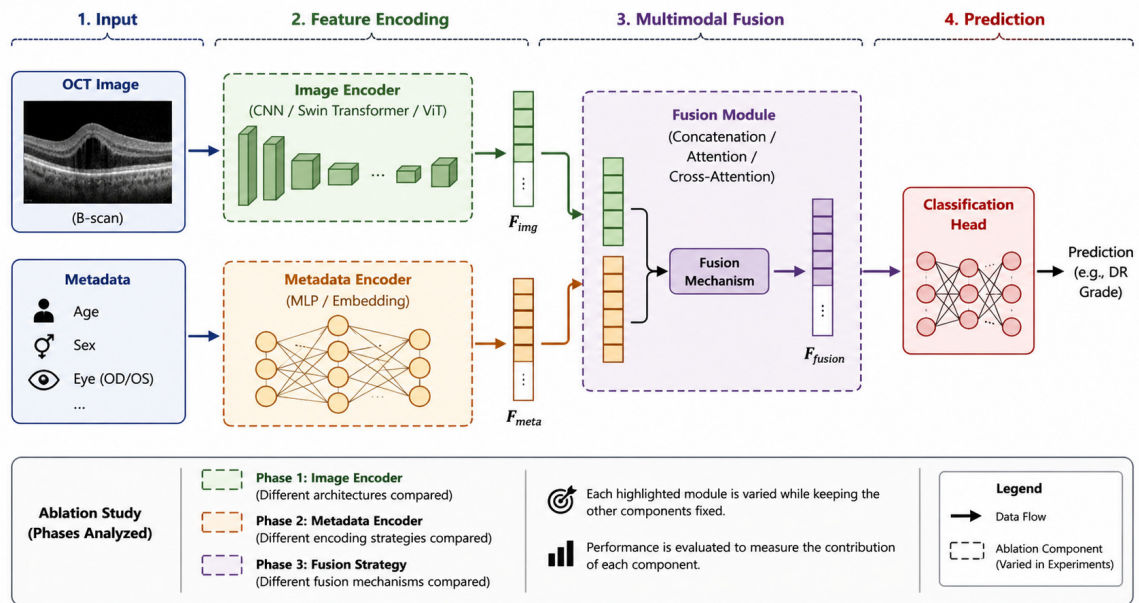


Figure 2.2: Overview of the proposed multimodal framework and ablation study design. The pipeline consists of two input branches (OCT images and clinical metadata) each processed by dedicated encoding modules. The resulting representations are integrated through alternative fusion strategies to produce the final diagnostic prediction. Dashed components indicate the modules evaluated during ablation.

2.7.1 Image Encoder Variants

Extracting discriminative visual representations from OCT B-scans is central to the proposed framework. OCT images contain fine-grained retinal microstructures and subtle pathological variations that require feature extractors capable of modeling both local spatial patterns and long-range structural dependencies. To assess the influence of encoder architecture, three representative families are evaluated: convolutional networks, flat vision transformers, and hierarchical transformers, instantiated as ResNet50, ViT, and Swin

Transformer, respectively.

All encoders are initialized with ImageNet pre-trained weights and fine-tuned on OCT data using transfer learning. In this phase, a [Multilayer Perceptron \(MLP\)](#) metadata encoder and a simple concatenation-based fusion strategy are held fixed across encoder comparisons to isolate the impact of the visual backbone.

Convolutional Baseline: ResNet50

ResNet50 ([He et al., 2016](#)) is employed as the convolutional baseline encoder. It learns hierarchical visual features through stacked convolutional layers equipped with residual skip connections, enabling stable optimization of deep architectures. In this work, ResNet50 is used to capture local retinal patterns including fluid accumulations, drusen deposits, and structural layer disruptions.

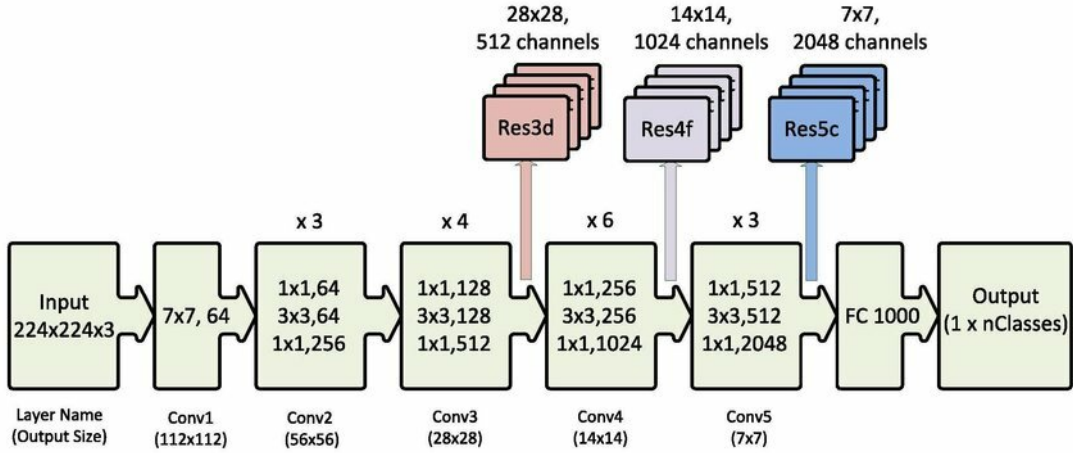


Figure 2.3: ResNet-50 architecture with residual connections ([Mahmood et al., 2016](#)).

Global Attention Encoder: Vision Transformer

ViT ([Dosovitskiy et al., 2020](#)) processes OCT images as sequences of fixed-size non-overlapping patches. Each patch is linearly projected into an embedding space and processed by a standard transformer encoder with global multi-head self-attention. ViT is included in the evaluation to assess global feature modeling in the absence of convolutional inductive bias.

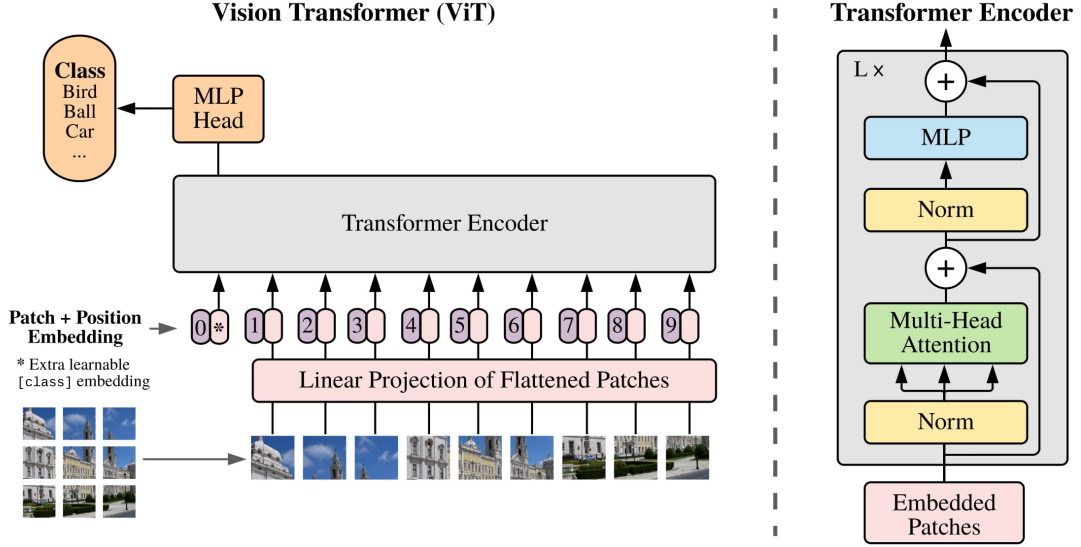


Figure 2.4: ViT pipeline: patch embedding and transformer encoder (Dosovitskiy et al., 2020).

Hierarchical Attention Encoder: Swin Transformer

Swin Transformer (Liu et al., 2021) introduces a hierarchical patch-based representation through shifted window self-attention. Local attention is computed within non-overlapping windows, while cross-window interaction is achieved through a shifted window partitioning scheme across alternating layers. This design combines the computational efficiency of local attention with the capacity for global contextual modeling through hierarchical feature aggregation.

Two variants are evaluated: Swin-Tiny (Swin-T) and Swin-Base (Swin-B), representing lightweight and high-capacity configurations, respectively.

2.7.2 Metadata Encoder Variants

Structured clinical metadata in this study comprises three variables: patient age (continuous), sex (binary categorical), and eye laterality (binary categorical). These low-dimensional tabular inputs must be transformed into a continuous embedding space before multimodal fusion. Two metadata encoding strategies are evaluated to assess representational capacity and their downstream impact on classification performance.

The **MLP encoder** processes the concatenated metadata vector through a sequence of fully connected layers with nonlinear activations, producing a compact dense embedding

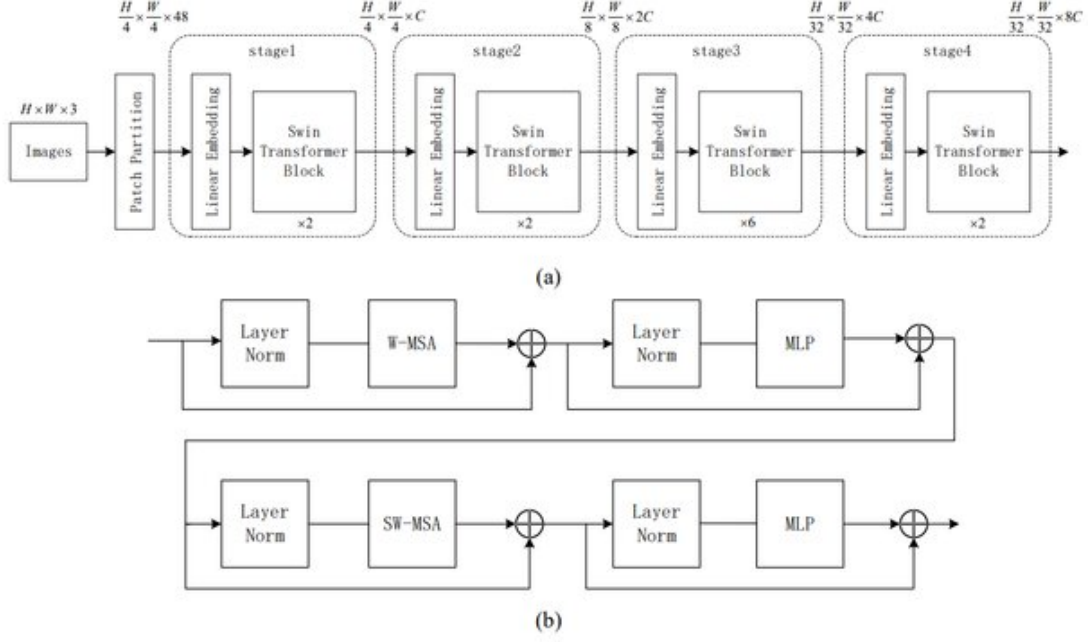


Figure 2.5: Swin Transformer with shifted window attention and hierarchical feature learning (Liu et al., 2021).

that captures feature interactions implicitly through parameter learning.

The **Transformer-based encoder** applies multi-head self-attention directly to the set of metadata variables, enabling explicit modeling of pairwise relationships and adaptive feature weighting even in low-dimensional input settings.

Both encoders are trained jointly with the Swin-B image encoder under identical experimental conditions, with a concatenation-based fusion strategy held fixed during this comparison phase.

2.7.3 Fusion Architectures

To evaluate multimodal integration strategies, several fusion mechanisms are investigated for combining image representations with clinical metadata embeddings. These approaches differ in the level of inter-modal interaction they support, ranging from simple feature aggregation to explicit cross-modal reasoning.

All fusion strategies are initially evaluated under a ResNet50 image encoder to establish a controlled comparison under a convolutional backbone. The best-performing strategy is subsequently validated using a Swin-B encoder to assess its generalizability

across architectural families.

Concatenation-Based Fusion. This baseline approach concatenates image and metadata embeddings into a single feature vector, which is then passed to a classification head. It is computationally efficient but does not explicitly model inter-modal interactions, relying instead on the classification head to learn cross-modal relationships implicitly.

Unidirectional Cross-Attention Fusion. This approach introduces asymmetric inter-modal interaction, where one modality attends over the other to generate a context-modulated representation. While it enables partial cross-modal conditioning, the interaction remains unidirectional and therefore does not fully exploit complementary dependencies between modalities.

Dual Cross-Attention Fusion (Proposed). To address the limitations of unidirectional interaction, a [DCA](#) mechanism is proposed to enable symmetric, bidirectional communication between [OCT](#) image features and clinical metadata embeddings.

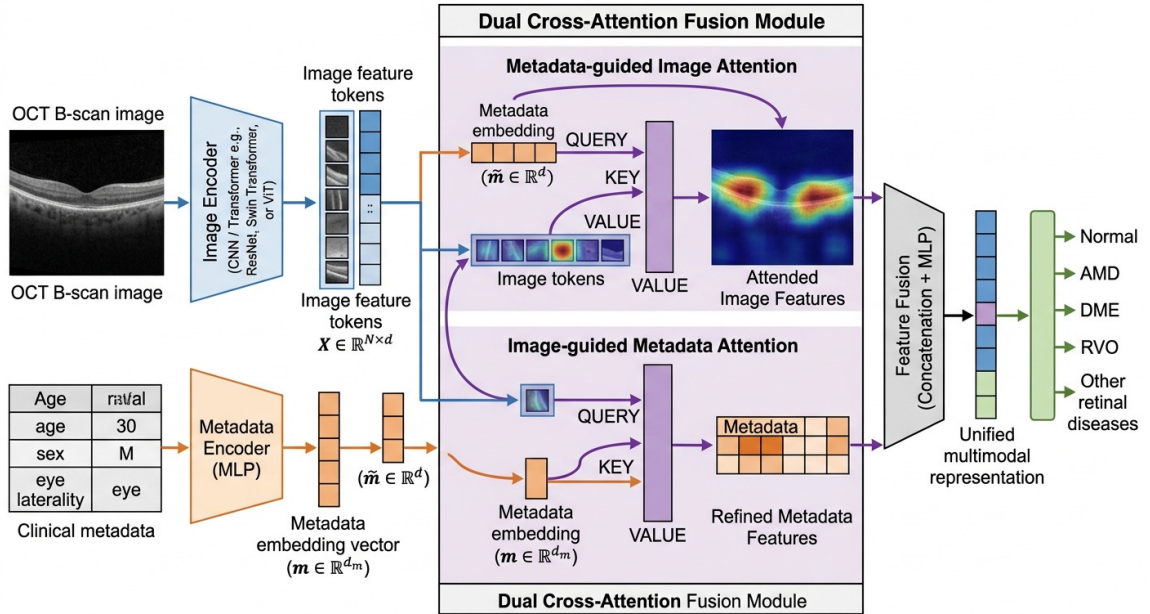


Figure 2.6: Dual Cross-Attention fusion mechanism enabling symmetric bidirectional interaction between [OCT](#) image tokens and clinical metadata embeddings.

Let $\mathbf{X} \in \mathbb{R}^{N \times d}$ denote the sequence of image feature tokens and $\mathbf{m} \in \mathbb{R}^{d_m}$ denote the metadata embedding vector. The metadata vector is first projected into the image

embedding space, producing $\tilde{\mathbf{m}} \in \mathbb{R}^d$.

The **DCA** mechanism then computes two complementary attention operations:

Metadata-to-Image Attention: Metadata-derived queries attend over image tokens to produce a metadata-conditioned visual representation, emphasizing retinal regions that are contextually relevant given the patient’s clinical profile.

Image-to-Metadata Attention: Image tokens attend over the metadata representation to produce an image-conditioned clinical embedding, refining the weighting of clinical features based on visual evidence.

The outputs of both attention directions are subsequently fused into a unified multimodal representation, which is passed to the classification head for final prediction.

2.7.4 Complete Multimodal Architecture

Following the ablation study of individual components, a complete multimodal architecture is defined for joint **OCT** and clinical representation learning within Framework I. The architecture integrates a Swin-B image encoder, an **MLP** metadata encoder, and a **DCA** fusion module augmented with a gated integration mechanism.

The Swin-B encoder extracts hierarchical visual representations from **OCT** B-scans through shifted window self-attention, capturing both local retinal microstructures and long-range spatial dependencies across the scan. The **MLP** metadata branch projects structured clinical variables "age, sex, and eye laterality" into a dense embedding space dimensionally compatible with the visual representation. The **DCA** module then models bidirectional inter-modal relationships, allowing each modality to dynamically condition its representation on complementary information from the other. A final gated fusion layer adaptively regulates the relative contribution of each modality, producing a unified representation for downstream classification.

2.7.5 Training and Optimization

The training procedure is designed to ensure stable optimization and robust model selection under class-imbalanced conditions.

- **Loss Function:** All experiments in Framework I are trained using weighted cross-entropy loss, with per-class weights inversely proportional to class frequency to mitigate the effect of class imbalance.
- **Optimizer:** Models are optimized using AdamW with a learning rate of 1×10^{-4} and weight decay of 1×10^{-4} , providing stable convergence with implicit regularization across encoder and fusion configurations.
- **Learning Rate Scheduling:** A cosine annealing scheduler is optionally applied to improve late-stage convergence behavior. [Automatic Mixed Precision \(AMP\)](#) Training is employed throughout to accelerate computation and reduce GPU memory usage without compromising numerical stability.
- **Model Selection:** Early stopping is applied based on macro-averaged F1-score on the validation set, which provides a reliable selection criterion under class imbalance. The checkpoint achieving the highest validation F1-score is retained as the final model for each experimental configuration.

2.8 Framework II: Clinical Decision Support Strategy

Framework II is designed independently from Framework I to address the fundamentally different characteristics of real-world clinical data. Unlike the controlled benchmark setting of Framework I, the clinical dataset involves weak patient-level supervision, incomplete and heterogeneous metadata, variable imaging conditions, and co-occurring pathologies. Consequently, architectural decisions in Framework II prioritize robustness, scalability, and deployment feasibility rather than ablation-driven multimodal optimization.

The framework is structured as an end-to-end hierarchical diagnostic pipeline comprising two sequential stages: a binary screening stage (Stage 1) to distinguish normal from abnormal [OCT](#) volumes, followed by a disease-specific multi-label classification stage (Stage 2) for pathological volumes. Both stages operate under a [GA](#) based Multiple In-

stance Learning (GA-MIL) formulation, motivated by the clinical reality that pathological signals in OCT volumes are often spatially localized to a subset of slices.

2.8.1 Multiple Instance Learning Formulation

Gated Attention-Based MIL Aggregation

Given a patient OCT volume represented as a bag of slices:

$$\mathcal{V}_i = \{x_{i1}, x_{i2}, \dots, x_{iT}\}$$

a frozen or fine-tuned encoder f_θ extracts a fixed-dimensional embedding for each slice:

$$h_{ij} = f_\theta(x_{ij})$$

These embeddings are aggregated into a patient-level representation using a gated attention mechanism:

$$\alpha_{ij} = \frac{\exp(\mathbf{w}^\top (\tanh(\mathbf{V}h_{ij}) \odot \sigma(\mathbf{U}h_{ij})))}{\sum_{k=1}^T \exp(\mathbf{w}^\top (\tanh(\mathbf{V}h_{ik}) \odot \sigma(\mathbf{U}h_{ik})))}$$

The patient-level representation is obtained as the attention-weighted sum of slice embeddings:

$$H_i = \sum_{j=1}^T \alpha_{ij} h_{ij}$$

The gating mechanism, implemented as an element-wise product of $\tanh$ and σ activations, selectively modulates slice-level features by amplifying diagnostically informative slices while suppressing non-informative or artifact-affected ones. This improves robustness to variability in image quality and acquisition noise.

Gated attention is preferred over simpler aggregation strategies such as mean or max pooling because it provides learnable, data-dependent weighting of slices rather than uniform or extremal aggregation. This is particularly beneficial in clinical OCT volumes, where informative slices are often sparse and interspersed with low-quality or irrelevant

ones.

Multi-Label Classification Formulation

Retinal diseases in the clinical dataset are not mutually exclusive; a single patient may concurrently present with conditions such as DR, AMD, and EDEMA. Stage 2 is therefore formulated as a multi-label learning problem, in which each pathology is predicted independently.

The classification head generates a vector of per-class logits:

$$\mathbf{z}_i = g_\phi(H_i) \in \mathbb{R}^K$$

where K denotes the number of target disease classes. Final class probabilities are obtained by applying an element-wise sigmoid transformation:

$$\hat{\mathbf{y}}_i = \sigma(\mathbf{z}_i) \in [0, 1]^K$$

Each pathology is modeled as an independent Bernoulli variable, enabling multiple co-occurring diagnoses. The Stage 2 training objective is the multi-label binary cross-entropy loss:

$$\mathcal{L}_{\text{Stage2}} = \text{BCEWithLogitsLoss}(\mathbf{z}_i, \mathbf{y}_i)$$

This formulation enables independent learning of each pathology while sharing the common latent representation H_i across all classification heads, allowing the model to capture both shared and disease-specific diagnostic patterns. Clinically, the attention weights α_{ij} also provide interpretability at the slice level, identifying which scans within a volume contribute most to the final diagnostic decision.

2.8.2 Feature Extraction Backbones

Each stage employs a dedicated encoder architecture selected to balance representational capacity with computational efficiency under the respective learning constraints.

- **Stage 1 — EfficientNet-B0:** A lightweight convolutional encoder based on MBConv blocks with squeeze-and-excitation modules. The model is initialized with ImageNet pre-trained weights and fine-tuned end-to-end on the clinical OCT data, producing 1,280-dimensional slice-level feature embeddings.
- **Stage 2 — SimCLR-Pretrained ResNet18:** A ResNet18 encoder pretrained under the SimCLR self-supervised contrastive learning framework on large-scale unlabeled data. In Stage 2, the encoder is used as a fixed feature extractor with frozen weights, preserving the self-supervised representations while only the GA-MIL module and the classification head are optimized during training. This design is motivated by the limited labeled data available at this stage, where end-to-end fine-tuning risks overfitting.

The SimCLR encoder was pretrained on the unlabeled portion of the clinical OCT dataset, allowing the self-supervised objective to learn retinal-specific low-level features prior to downstream adaptation.

EfficientNet-B0 was not adopted for Stage 2 because the limited availability of fine-grained labeled data at this stage makes end-to-end fine-tuning prone to overfitting; the SimCLR-pretrained ResNet18 encoder, by contrast, provides expressive representations without requiring label-driven adaptation of the backbone.

2.8.3 Training and Optimization

- **Loss Functions:** Stage 1 employs a weighted Binary Cross-Entropy loss to account for class imbalance in the binary screening task. Stage 2 uses BCEWithLogitsLoss applied to patient-level predictions from the GA-MIL framework, enabling multi-label disease classification under weak supervision.
- **Optimizer and Scheduling:** Both stages use the AdamW optimizer with a learning rate of 1×10^{-4} and weight decay of 1×10^{-4} . A cosine annealing learning rate scheduler is applied in Stage 1 to improve convergence stability. AMP training

is employed in both stages to accelerate computation and reduce GPU memory consumption.

2.9 Conclusion

This chapter presented the methodological design developed for automated macular disease diagnosis using OCT imaging and structured clinical metadata. The proposed approach is organized into two complementary frameworks, each addressing a distinct operational setting and learning objective.

Framework I introduced a research-oriented experimental setting based on a publicly available OCT benchmark dataset. It was designed to systematically investigate multimodal representation learning under controlled conditions. This framework enabled a detailed analysis of different image encoders, metadata encoders, and fusion strategies through structured ablation studies, with particular emphasis on transformer-based architectures and the proposed dual cross-attention mechanism for enhanced multimodal interaction.

Framework II focused on a clinically driven decision support strategy built on a real-world dataset collected at the Ibn Al Haythem Ophthalmology Center. Unlike the benchmark setting, this framework was designed to operate under realistic clinical constraints, including weak supervision, heterogeneous imaging conditions, and incomplete metadata. A hierarchical diagnostic pipeline based on multiple instance learning was introduced to support patient-level prediction and multi-label disease classification.

Overall, the proposed methodology establishes an experimental and clinical pipelines for OCT-based retinal disease analysis. It bridges controlled benchmarking with real-world deployment requirements, while systematically studying the impact of multimodal learning, attention-based fusion mechanisms, and weakly supervised learning in ophthalmic diagnosis. The following chapter presents the experimental results.

Chapter 3

Results and Discussion

3.1 Introduction

This chapter presents the experimental results of the proposed multimodal clinical decision support system for retinal disease diagnosis using OCT images and patient metadata. The evaluation is conducted in two stages: a controlled ablation study on a public dataset and a real-world clinical validation dataset. The performance is analyzed using multiple metrics including Accuracy (Acc), ROC-AUC, F1-score, Precision-Recall Area Under the Curve (PR-AUC), sensitivity, and specificity, in addition to qualitative explainability analysis.

3.2 Framework I: Experimental Multimodal Learning Models Evaluation

This section presents an ablation study on the benchmark dataset under a controlled experimental setup. All models are initialized with ImageNet pre-trained weights and trained using the same patient-wise data split and identical optimization configuration to ensure a fair comparison across architectures.

The evaluation is performed on the same held-out test set using identical metrics. The analysis is structured progressively, beginning with a comparison between unimodal and multimodal models, followed by an assessment of different image encoders, metadata

encoding strategies, and fusion mechanisms. Finally, the performance of the proposed dual cross-attention model is examined in detail.

3.2.1 Comparative Analysis of Metadata Recovery Approaches

This subsection evaluates the performance of the metadata recovery module for estimating age, sex, and eye laterality from OCT B-scans. Two learning strategies are compared: the Task-Specific Framework (TSF), which trains independent models per task, and the Multi-Head Framework (MHF), which jointly learns all tasks using a shared backbone.

Overall Results. Table 3.1 summarizes the best-performing configurations. TSF consistently outperforms MHF across all attributes. The strongest results are obtained using Swin-B for age regression (MAE = 4.27 years) and eye laterality classification (Accuracy = 0.933), while a hybrid DenseNet-121 + Transformer achieves the best sex classification performance (Accuracy = 0.846).

Table 3.1: Summary of best metadata recovery performance for TSF and MHF.

Task	Best TSF Performance	Best MHF Performance
Age (MAE ↓)	4.27 (Swin-B)	6.45 (Swin-Tiny)
Sex (Accuracy ↑)	0.846 (DenseNet+T)	0.779 (ResNet-50)
Eye laterality (Accuracy ↑)	0.933 (Swin-B)	0.761 (ResNet-50)

Discussion. The performance gap between TSF and MHF is mainly attributed to gradient interference in multi-task optimization, where regression and classification objectives compete within a shared representation space. In contrast, TSF benefits from task-specific specialization and higher-capacity encoders, leading to more accurate recovery of clinical metadata at the cost of increased computational complexity.

Overall, although MHF offers better efficiency and a unified design, TSF provides significantly more accurate metadata reconstruction, which is crucial for reducing semantic inconsistencies in the OCTDL dataset.

3.2.2 Unimodal vs Multimodal Performance

This section evaluates the performance difference between the unimodal OCT-only model and the proposed multimodal model that integrates OCT features with patient metadata (age, sex, and eye laterality).

Table 3.2 presents the quantitative comparison across key evaluation metrics. The results show that the multimodal model consistently outperforms the unimodal baseline, with improvements in Acc, F1-score, sensitivity, and specificity.

Model	Acc	F1-score	Sensitivity	Specificity
OCT Only	0.8933	0.8927	0.8412	0.9784
OCT + Metadata	0.9107	0.9103	0.8636	0.9840

Table 3.2: Performance comparison between unimodal and multimodal models using the OCTDL dataset with a ResNet-50 encoder on the test set.

The unimodal ResNet-50 model achieves strong performance by learning discriminative features directly from OCT images, particularly for well-defined classes such as NO and AMD. However, its performance decreases for visually similar or subtle conditions such as ERM and RVO, reflecting limitations in capturing fine-grained variations, which is also reflected in its lower sensitivity (0.8412).

The multimodal model improves overall performance across all metrics, with the most notable gain in sensitivity (0.8636), reducing false negatives and improving disease detection reliability. The increase in specificity (0.9840) also indicates better discrimination of healthy cases.

These improvements are mainly attributed to the complementary role of metadata, which provides additional patient-level context not directly observable in OCT images, helping to better distinguish ambiguous cases such as ERM and RVO.

Performance gains are less pronounced in underrepresented classes such as VID and RAO, likely due to limited sample sizes and resulting statistical variability.

Overall, the results confirm that integrating patient metadata with OCT imaging improves diagnostic performance, particularly by enhancing sensitivity and robustness in challenging cases.

3.2.3 Image Encoding Analysis

To isolate the effect of image encoders, all experiments use a fixed multimodal fusion strategy based on feature concatenation, while patient metadata is encoded using a MLP. This ensures that performance variations are mainly attributable to the image encoding backbone.

Results of Image Encoding Architectures

Model	Val Acc	Test Acc
ResNet50 + Metadata	0.9279	0.9107
Swin-T + Metadata	0.9423	0.9082
Swin-B + Metadata	0.9447	0.9330
ViT + Metadata	0.9038	0.8834

Table 3.3: Performance comparison of image encoding architectures on the OCTDL

Table 3.3 shows clear performance differences between convolutional and transformer-based architectures. ResNet50 achieves competitive results, confirming the effectiveness of convolutional feature extraction for OCT analysis. Both Swin Transformer variants outperform the remaining models, with Swin-B achieving the highest test Acc (0.9330). In contrast, ViT shows the lowest test performance (0.8834), despite satisfactory validation behavior.

Role of Architectural Inductive Bias in OCT Feature Learning

The observed performance differences can be interpreted through the inductive biases of the evaluated architectures, which influence how spatial relationships are modeled in OCT images. Figure 3.1 illustrates the conceptual differences between CNN-based, transformer-based, and hierarchical transformer approaches.

CNNs rely on local receptive fields, making them effective for capturing fine retinal structures such as layer boundaries and localized lesions, but less effective at modeling long-range dependencies. In contrast, Vision Transformers use global self-attention, enabling flexible contextual modeling but often requiring larger datasets for stable generalization.

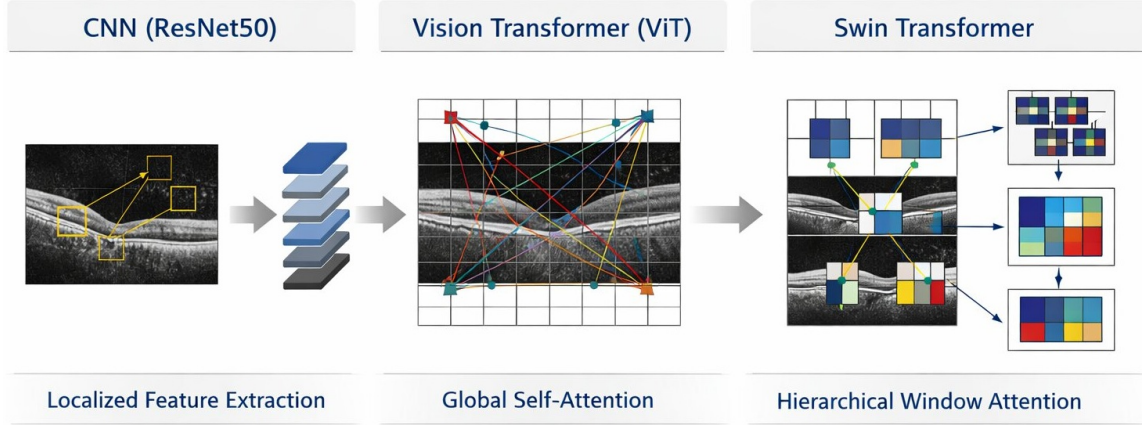


Figure 3.1: Comparison of inductive biases in CNN, ViT, and Swin Transformer for OCT representation learning.

Hierarchical transformers such as Swin-B combine local window attention with cross-window interactions, enabling simultaneous modeling of local anatomical details and broader retinal context. This balanced inductive bias likely explains the superior performance of Swin-B observed in the experiments.

3.2.4 Clinical Data Encoder Analysis

The performance of the two metadata encoding strategies is summarized in Table 3.4. During training, OCT B-scan images are processed through the Swin-B backbone, while structured clinical variables (age, sex, and eye laterality) are encoded separately using either an MLP-based encoder or a Transformer-based encoder. The resulting metadata embedding is fused with the visual representation before classification.

Metric	Swin-B + MLP Encoding	Swin-B + Transformer Encoding
Val Acc	0.9471	0.9399
Test Acc	0.9330	0.9256

Table 3.4: Comparison of metadata encoding strategies on the OCTDL

The results show that the choice of metadata encoder has a modest but consistent impact on performance, with the MLP-based encoder achieving slightly higher validation and test Acc.

This behavior can be attributed to the nature of the clinical variables, which are low-dimensional and do not exhibit sequential or hierarchical structure. In this context,

simple nonlinear transformations provided by an [MLP](#) are sufficient to generate effective feature representations for multimodal fusion.

In contrast, the Transformer-based encoder does not provide additional gains and slightly underperforms. Although self-attention mechanisms are powerful for modeling complex relationships, they are less suitable for small tabular inputs where feature interactions are limited, leading to unnecessary model complexity without improved representation quality.

These observations suggest that the effectiveness of metadata encoding strategies is strongly dependent on the structure and dimensionality of the input features. For low-dimensional clinical data, lightweight encoders such as [MLPs](#) are more appropriate, whereas attention-based models may be more beneficial when richer and more structured metadata is available.

Overall, the results indicate that [MLP](#)-based encoding provides a more efficient and well-suited solution for integrating simple clinical variables in multimodal [OCT](#) classification frameworks.

3.2.5 Fusion Strategy Analysis

This section evaluates three multimodal fusion strategies under a fixed experimental setup using ResNet50 for image encoding and [MLP](#) for metadata encoding. Performance differences across models are therefore attributable solely to the fusion mechanism. The analysis encompasses three complementary dimensions: global classification performance, calibration behavior, and class-wise sensitivity.

Global Performance

The Concatenation-based model achieves the highest [ROC-AUC](#) (0.9846), indicating strong global ranking capability, but this does not translate into well-calibrated or balanced class performance. In contrast, the Dual Cross-Attention model achieves the highest macro F1-score (0.8746) and the lowest Expected Calibration Error ([ECE](#) = 0.0592), demonstrating a more favorable balance between discriminative [Acc](#) and prediction reli-

Metric	Concatenation	Cross-Attn.	Meta→Image	Image→Meta
F1 Macro	0.8212	0.8746	0.8054	~0.60
F1 Weighted	0.9308	0.9099	0.9161	~0.60
Precision	0.8584	0.8923	0.8207	~0.60
Recall	0.8125	0.8674	0.8166	~0.60
ROC-AUC	0.9846	0.9648	–	low
ECE (↓)	0.2600	0.0592	0.0812	high
Inference Time	2.30s	0.85s	1.47s	unstable

Table 3.5: Performance comparison of fusion strategies for the OCTDL on the test set ability.

The Image-to-Metadata attention variant degrades to near-random performance, confirming that the imaging modality plays a substantially more dominant role in this task and that unidirectional attention from image to metadata is unable to produce meaningful multimodal representations.

Calibration and Clinical Reliability

Model	ECE	Minority Class Performance	Clinical Usability
Concatenation-Based	0.260	Weak	Unsuitable
Dual Cross-Attention	0.059	Strong	Deployable
Meta→Image	0.081	Weak	Limited
Image→Meta	High	Very weak	Unusable

Table 3.6: Evaluation of the clinical reliability of multimodal fusion strategies for OCTDL dataset on the test set

Calibration quality is a critical differentiator among fusion strategies. Despite its strong discriminative performance, the Concatenation-based model is substantially overconfident (ECE = 0.260), which limits its safety in clinical decision support where miscalibrated confidence scores can lead to inappropriate risk assessment. The Dual Cross-Attention model significantly reduces this issue, producing well-calibrated probability estimates aligned with true outcome frequencies.

Class-Wise Sensitivity Analysis

The Dual Cross-Attention model provides the most balanced sensitivity across disease categories, with particularly notable improvements in the structurally challenging Class 2 and perfect recall in Class 3. The Meta→Image variant exhibits a severe recall collapse

Class	Concat.	Cross-Attn.	Meta→Image	Image→Meta
0	~0.94	0.940	0.90	~0.60
1	~0.85	0.854	0.80	~0.60
2	~0.82	0.941	0.44	~0.60
3	~0.98	1.000	0.98	~0.60
4	1.000	1.000	1.00	~0.60
5	~0.88	0.860	0.84	~0.60
6	~0.68	0.725	0.75	~0.60

Table 3.7: Comparison of class-wise recall across different multimodal fusion strategies for the OCTDL dataset on the test set

for Class 2 (0.44), likely attributable to asymmetric attention failing to exploit imaging features effectively for visually ambiguous pathologies.

Discussion

The results reveal a fundamental trade-off between discriminative performance and calibration reliability across fusion strategies. Concatenation-based fusion maximizes ranking performance but sacrifices calibration and balanced class sensitivity. Unidirectional attention mechanisms provide limited advantages and are highly sensitive to modality dominance asymmetry.

The Dual Cross-Attention architecture consistently achieves the most clinically appropriate trade-off. Its symmetric bidirectional design enables both modalities to iteratively condition and refine each other’s representations, producing improvements in macro F1-score, minority class sensitivity, and calibration quality simultaneously. From a clinical deployment perspective, reliable uncertainty quantification and robust performance across all disease categories are of greater practical importance than marginal gains in global [ROC-AUC](#). The Dual Cross-Attention mechanism is therefore selected as the fusion strategy for the complete proposed architecture.

3.2.6 Impact of Image Encoder Backbone within the Fusion Framework

Although previous experiments identified Swin-B as the strongest image encoder, an additional comparison with ResNet50 is conducted within the proposed dual Cross-Attention fusion framework to evaluate whether the performance gains remain consistent under

multimodal fusion.

Table 3.8 presents the global performance comparison. Swin-B maintains superior performance across all evaluation metrics, with the largest improvements observed in recall and macro F1-score, indicating better sensitivity and robustness for multimodal classification.

Metric	ResNet50	Swin-B	Improvement
Macro F1-score	0.8746	0.9052	+0.0306
Weighted F1-score	0.9099	0.9291	+0.0192
Precision	0.8923	0.9342	+0.0419
Recall	0.8674	0.9280	+0.0606

Table 3.8: Performance comparison of ResNet-50 and Swin-B encoders within a dual cross-attention fusion framework for the OCTDL dataset on the test set

Class-wise F1-scores in Table 3.9 show that the improvements are particularly noticeable in challenging and minority classes, especially Classes 5 and 6.

Model	Class 0	Class 1	Class 2	Class 3	Class 5	Class 6
ResNet50	0.949	0.875	0.901	0.889	0.818	0.690
Swin-B	0.958	0.914	0.899	0.908	0.894	0.765

Table 3.9: Comparison of class-wise F1-scores between ResNet-50 and Swin-B encoders within a dual cross-attention fusion framework for the OCTDL dataset on the test set

Error analysis indicates that Swin-B reduces inter-class confusion observed with ResNet50, particularly for visually similar pathological categories. Calibration analysis in Table 3.10 further shows lower Expected Calibration Error and improved confidence separation, suggesting more reliable probability estimation.

Metric	ResNet50	Swin-B
ECE (Before $\rightarrow$ After)	0.0875 $\rightarrow$ 0.0656	0.0537 $\rightarrow$ 0.0291
Temperature Stability	Lower	Higher
Confidence Separation	Weaker	Stronger

Table 3.10: Comparison of predictive calibration performance between ResNet-50 and Swin-B encoders in the multimodal OCTDL classification framework on the test set.

These results confirm that the advantages of Swin-B remain consistent within the multimodal fusion framework, leading to improved robustness, calibration quality, and minority-class recognition.

3.2.7 Final Swin-B DCA Model

Based on the complete ablation study, the final proposed model combines a Swin-B image encoder with a dual cross-attention fusion mechanism. This configuration was selected because it consistently achieved the best balance between classification performance, multimodal integration, and calibration reliability across all evaluated experiments.

Overall, the proposed model demonstrates strong and well-balanced multi-class performance, achieving a macro F1-score above 0.90 despite class imbalance within the dataset. Weighted metrics further confirm high predictive performance on dominant categories while maintaining stable behavior on more challenging pathological classes.

At the class level, the model achieves consistently high performance for major categories (0–3), with particularly strong separability for Class 3. Although performance decreases slightly for minority and visually similar classes, especially Class 6, the model maintains stable classification behavior without critical degradation.

In addition to classification [Acc](#), the model provides improved confidence calibration and uncertainty estimation, reflected by reduced [ECE](#) and clearer confidence separation between correct and incorrect predictions. These properties are important for reliable clinical decision support.

Overall, the experimental results confirm that the Swin-B dual cross-attention architecture provides the most effective and reliable configuration among all evaluated models for [OCT](#)-based retinal disease classification.

3.2.8 Explainability Analysis

Visual Explainability via Grad-CAM

[Grad-CAM](#) heatmaps are generated by extracting intermediate feature activations and gradients from the Swin-B encoder and projecting them onto the input [OCT](#) images. Representative visualizations are presented in [Figure 3.2](#).

Across correctly classified samples, model attention consistently localizes to clinically meaningful retinal structures, including fluid accumulation regions, structural layer dis-

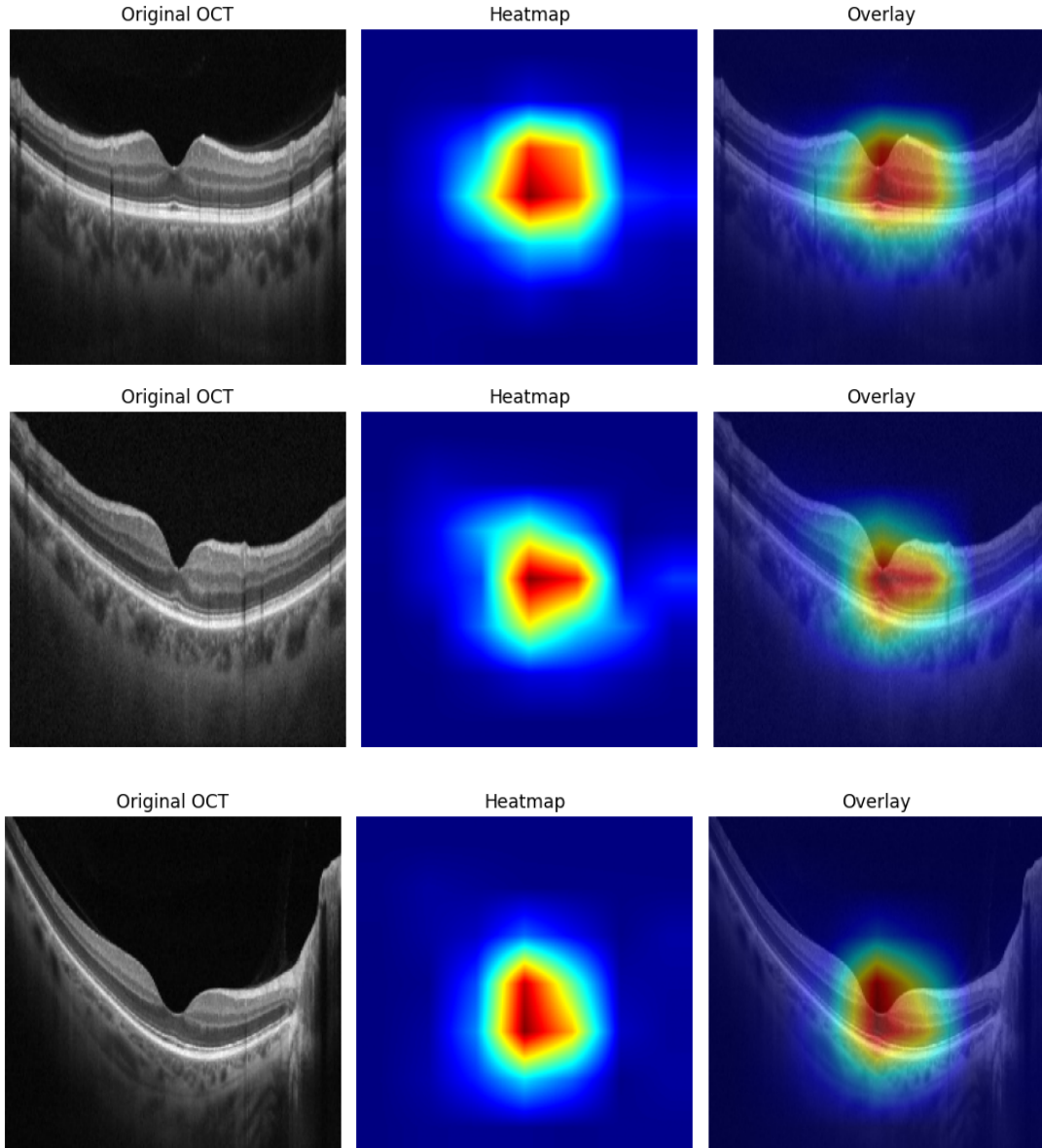


Figure 3.2: [Grad-CAM](#) visualization triplets: original [OCT](#) image, activation heatmap, and overlay.

ruptions, and pathological reflectivity changes, rather than image artifacts or background regions. This spatial alignment confirms that the model’s predictions are driven by anatomically relevant features.

Failure Mode Analysis. Analysis of the 29 misclassified samples reveals that the large majority (89.7%) are associated with diffuse, unfocused attention patterns indicating the absence of a dominant discriminative region, rather than with systematic misfocusing or background attention. This suggests that most classification errors arise from intrinsic

Failure Type	Count	Percentage (%)
Diffuse Attention Failures	26	89.7
Edge/Border Failures	3	10.3
Total	29	100

Table 3.11: Analysis of misclassified samples using Grad-CAM for the [DCA](#) framework for OCTDL on the test set

visual ambiguity in the input images rather than from model-specific biases or representation failures.

Robustness Analysis under Input Perturbations The model is evaluated under simulated clinical degradations without retraining:

Metric	Original	Gaussian Noise	Gaussian Blur	Brightness Drop
Acc	0.9280	0.9181	0.8288	0.9256

Table 3.12: Robustness evaluation under input perturbations in the proposed multimodal OCTDL classification framework on the test set.

The model is highly robust to noise and brightness variations, with minimal degradation. However, Gaussian blur causes a more significant drop (-0.0992), reflecting sensitivity to loss of fine structural details essential in [OCT](#) imaging.

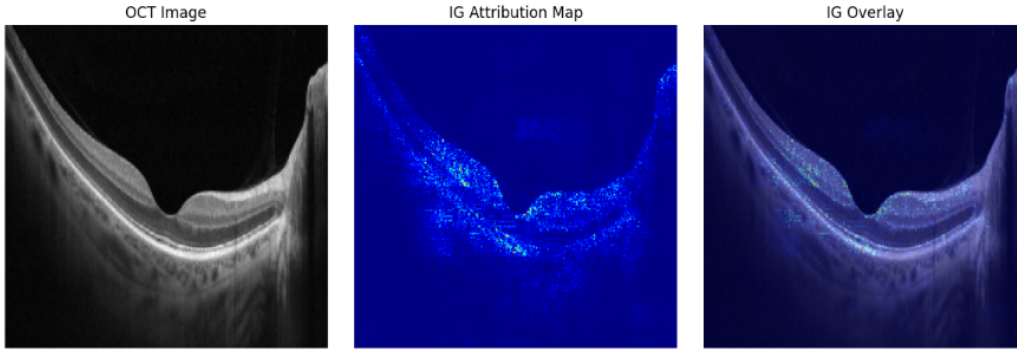
Despite this, overall performance remains strong, confirming good robustness under realistic clinical conditions.

Multimodal Attribution via Integrated Gradients

Integrated Gradients are applied independently to the image and metadata branches of the multimodal architecture to quantify per-feature attribution scores.

Pixel-level attribution maps confirm that the visual branch assigns strong positive attributions to pathologically relevant retinal regions, including areas of structural deformation, intraretinal or subretinal fluid, and abnormal reflectivity patterns, while background regions and acquisition artifacts receive near-zero attributions.

For the metadata branch, patient age consistently yields the highest attribution scores across pathological categories, consistent with the known age-dependent prevalence of macular diseases such as [AMD](#). In contrast, sex and eye laterality exhibit comparatively



```
{'prediction': 3,
 'class_name': 'NO',
 'confidence': 0.9179291725158691,
 'delta': -0.18816423416137695,
 'meta_attributions': {'age': 0.19625863432884216,
 'sex': -5.428920849226415e-05,
 'eye': 0.0}}
```

Figure 3.3: Integrated Gradients attribution analysis. Top: pixel-level attribution maps from the OCT image branch. Bottom: feature-level attribution scores for structured clinical metadata variables.

lower attribution magnitudes, reflecting their more limited discriminative contribution under the current dataset distribution. These findings confirm that the multimodal framework performs selective and clinically coherent feature integration across modalities.

Anatomically Guided Retinal Layer Attribution

To provide clinically interpretable spatial grounding, Grad-CAM activation maps are spatially aligned with retinal layer segmentation masks generated by a DeepLabV3-ResNet50 model, enabling layer-wise attribution quantification.

In pathological cases, peak activation responses are consistently localized within inner and outer retinal layers associated with structural disruption, fluid accumulation, and abnormal reflectivity, structures that are routinely examined by clinicians during macular disease diagnosis. The spatial correspondence between model attention and anatomically meaningful retinal regions provides evidence of clinical plausibility and supports the interpretability of the proposed framework for ophthalmological deployment.

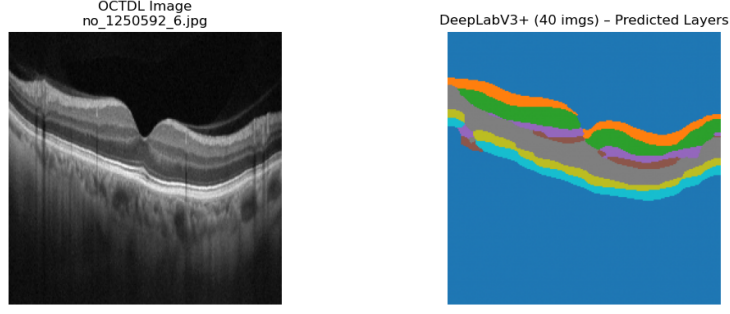


Figure 3.4: Retinal layer segmentation generated by the DeepLabV3-ResNet50 model, identifying seven anatomically distinct retinal layers.

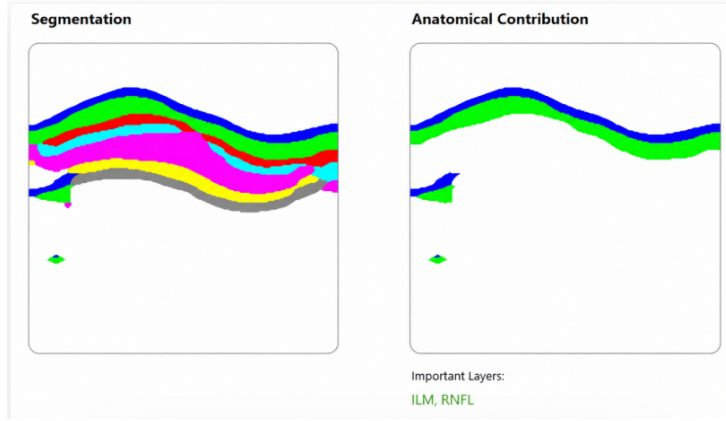


Figure 3.5: Anatomically guided explainability overlay combining Grad-CAM activation maps with retinal layer segmentation masks.

3.2.9 Comparison with State-of-the-Art Methods

To contextualize the proposed framework, its performance is compared against published methods on the OCTDL benchmark. Methods are grouped by architecture family and training setting; all reported results correspond to the seven-class OCTDL classification task.

Method	Architecture	Acc	W-F1	Macro F1	Setting
<i>CNN-based Methods</i>					
Kulyabin et al. (Kulyabin et al., 2023)	VGG16	0.8810	0.852	–	OCTDL only
Kulyabin et al. (Kulyabin et al., 2023)	ResNet50	0.8820	0.857	–	OCTDL only
<i>Transformer-based Methods</i>					
Hossain et al. (Khan et al., 2023)	SwinVFTR	0.9210	0.912	–	OCTDL only
Li et al. (He et al., 2023)	Swin-Poly Transformer	0.9380	–	–	OCT2017 + OCT-C8
Zhang et al. (Zhang et al., 2026)	ViT-Base	0.9361	–	–	OCTDL + external data
<i>Proposed Method</i>					
Ours	Swin-B + Dual Cross-Attention	0.9350	0.9291	0.9052	OCTDL only + metadata

Table 3.13: Comparison with state-of-the-art methods on the OCTDL benchmark. Results are grouped by architecture family and training setting.

Discussion. CNN-based methods achieve accuracies around 88%, reflecting the limitations of local convolutional feature extraction for fine-grained retinal pathology recognition. In contrast, transformer-based architectures improve performance, with Swin-based models reaching 92–93.8%, benefiting from hierarchical attention that better captures spatially distributed pathological patterns.

On the OCTDL dataset, the proposed model achieves 93.50% [Acc](#), a weighted F1-score of 0.9291, and a macro F1-score of 0.9052, placing it among the top-performing approaches. Its [Acc](#) is slightly below Swin-Poly (93.80%), which additionally uses OCT2017 and OCT-C8 for training, making the comparison not strictly equivalent. Similarly, the ViT-Base approach of Zhang et al. relies on external data augmentation.

Importantly, the proposed method is the only approach evaluated exclusively on OCTDL that integrates structured clinical metadata via a dual cross-attention mechanism. Despite this stricter setting, it maintains competitive [Acc](#) and achieves a strong macro F1-score, which is particularly relevant under class imbalance where it provides a more reliable measure of performance than [Acc](#) alone.

3.3 Framework II: Two-Stage Clinical Decision Support Model

Evaluation

This section evaluates the proposed two-stage framework for automated [OCT](#) analysis using data from the Ibn Al Haytham Center. The framework operates hierarchically: a binary screening stage distinguishes normal from abnormal cases, followed by a fine-grained disease-specific classification stage.

The Ibn Al Haytham Center dataset is used for training and internal validation, while the OCTDL dataset serves exclusively as an independent external validation set and is not involved in any stage of model development.

To ensure an optimal framework configuration, preliminary experiments were conducted to analyze the impact of slice selection, aggregation strategies, and multimodal fusion on model performance.

Preliminary Experiments and Design Choices

All experiments are conducted at either slice-level or patient-level depending on the aggregation strategy, with slice-level models processing individual OCT slices and patient-level models aggregating multiple slices within a patient.

Key findings from the preliminary experiments guided the final design: central macular slices consistently provide the most reliable diagnostic information; attention-based aggregation outperforms simple pooling by suppressing non-informative slices; and meta-data fusion via simple concatenation does not yield consistent improvements, suggesting that OCT imaging remains the dominant predictive modality.

These findings are supported by the results in Table 3.14. The baseline ResNet18 achieves ROC-AUC = 0.7651, but patient-level pooling over all four slices degrades performance (ROC-AUC = 0.6611), confirming that uniform aggregation introduces noise. Early OCT-metadata fusion further reduces performance (ROC-AUC = 0.5556). Attention over all four slices partially recovers this (ROC-AUC = 0.6833), while restricting input to central slices with EfficientNet and attention yields the best preliminary result (ROC-AUC = **0.8056**).

Model	Input Strategy	Best ROC-AUC
ResNet18 Baseline	Single OCT slice	0.7651
ResNet18 + Pooling	All 4 slices (aggregation)	0.6611
ResNet18 + Fusion	OCT + Metadata	0.5556
ResNet18 + Attention	Attention over 4 slices	0.6833
EfficientNet + Attention	All 4 slices	0.7000
EfficientNet + Attention	Selected middle slices	0.8056
EfficientNet + Safe Fusion	OCT + Metadata (stable fusion variants)	0.7000

Table 3.14: results of the ablation study for Stage 1 models on the test set.

This finding motivated expanding the slice extraction window (approximately indices 120–150) for broader macular and retinal coverage, which yielded further performance gains. Label noise was also identified as a potential source of variability.

Overall, these results guided the final architectural and preprocessing decisions, highlighting the importance of slice selection, the limitations of naive aggregation and fusion strategies, and the impact of data quality on model performance.

3.3.1 Stage 1: Binary Classification (Screening)

The first stage performs patient-level binary classification to distinguish normal from abnormal OCT cases. Three progressively refined MIL architectures were evaluated, with analysis covering classification performance, calibration quality, interpretability, and external generalization.

Architectural Evolution of the MIL Framework

Table 3.3.1 summarizes the key design changes across the three models. Models A and B rely on a ResNet18 backbone (512-D embeddings) with basic and improved attention, respectively. Model C adopts EfficientNet-B0 (1280-D embeddings) paired with a gated attention mechanism, AdamW optimization with cosine annealing, mixed precision training, and weighted random sampling for class imbalance handling. These modifications collectively improve convergence stability, reduce overfitting, and enhance generalization.

Component	Model A	Model B	Model C
Backbone	ResNet18	ResNet18	EfficientNet-B0
Feature dimension	512	512	1280
Attention mechanism	Simple attention	Improved weighted attention	Gated attention
Optimizer	Adam	Adam	AdamW + cosine annealing
AMP	No	No	Yes
Class imbalance handling	Positive weighting	Positive weighting + sampling	Weighted random sampler
Training stability	Low	Moderate	High

Table 3.15: Architectural comparison of the evaluated MIL models.

Comparative Performance Evaluation

Table 3.16 reports classification performance on the held-out test set. Model A achieves a reasonable ROC-AUC (0.92) but exhibits low sensitivity (0.58), indicating a high false-negative rate. Model B improves across all metrics through better aggregation, augmenta-

tion, and imbalance handling, though it remains constrained by the ResNet18 backbone. Model C achieves the best overall performance, with 96.88% **Acc**, 0.9875 **ROC-AUC**, and 96.49% sensitivity while maintaining perfect specificity, driven by the EfficientNet-B0 backbone and gated attention mechanism. Clinically, high sensitivity minimizes missed pathological cases while

The consistently high specificity suggests that the decision boundary is conservative with respect to normal cases, which is desirable in screening systems to minimize unnecessary clinical escalation.

Model	Acc	F1-score	ROC-AUC	Sensitivity	Specificity
Model A	0.62	0.73	0.92	0.58	1.0000
Model B	0.84	0.90	0.977	0.82	1.0000
Model C	0.9688	0.9821	0.9875	0.9649	1.0000

Table 3.16: Performance comparison of the evaluated models for the real-world test dataset on the test set.

Calibration and Reliability Analysis

Table 3.17 reports calibration performance. Model A produces poorly calibrated probabilities (**ECE** = 0.386), while Model B reduces calibration error substantially. Model C achieves the best calibration (Brier Score = 0.085, **ECE** = 0.159), ensuring that predicted probabilities are well aligned with true clinical outcomes is a critical property for clinical decision support.

Metric	Model A	Model B	Model C
Brier Score	0.247	0.115	0.085
ECE	0.386	0.194	0.159

Table 3.17: Comparison of predictive calibration performance across the evaluated models during the real-world screening stage of the **OCT** classification framework.

Attention, Confidence, and Interpretability Analysis

Model C exhibits strong statistical separation between normal and abnormal cases: normal cases have a mean confidence of 0.0600 versus 0.6803 for abnormal cases ($p <$

0.000001). Attention entropy is lower for normal cases (2.5734) than for abnormal cases (2.7548, $p = 0.000004$), indicating that pathological cases distribute diagnostic information across multiple slices rather than concentrating it in a single region. Accordingly, normal cases show higher maximum attention concentration (0.1434 vs. 0.0796, $p < 0.000001$), reflecting more localized focus in healthy scans. A significant negative correlation between attention entropy and confidence ($r = -0.4704$, $p = 0.0001$) confirms that attention behavior is meaningfully linked to prediction certainty.

Top-attention slice analysis confirms that the model consistently prioritizes diagnostically relevant regions, with significant differences between normal and abnormal cases ($p = 0.000241$) and between random and top-attention slices ($p < 0.000001$ for abnormal cases), validating that the gated attention mechanism focuses on clinically meaningful structures.

Grad-CAM Visual Interpretability Analysis

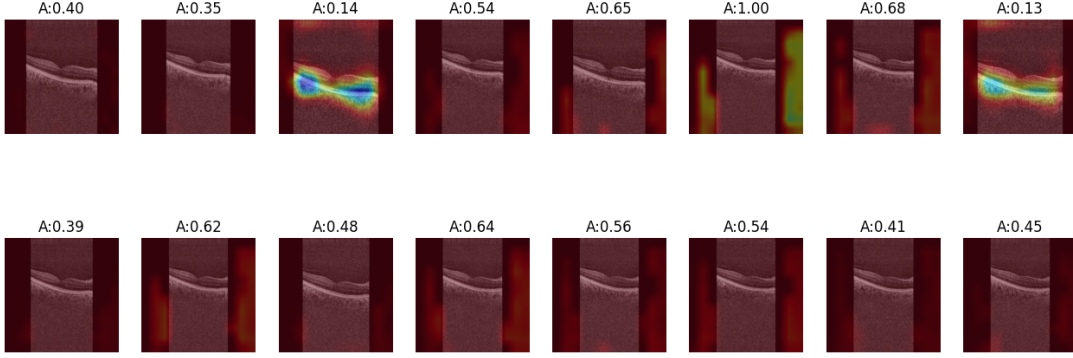
Grad-CAM visualizations combined with MIL attention confirm that the model learns clinically meaningful spatial representations. In abnormal cases, activations are consistently concentrated in central retinal layers across multiple slices, regions associated with fluid accumulation and structural distortion. In normal cases, activations are weak and diffuse, often outside retinal structures, reflecting the absence of pathological signals (Figure 3.6).

External Validation and Generalization

Table 3.18 compares internal and external validation performance. Despite domain shift in imaging devices, populations, and annotation protocols, Model C maintains strong discrimination externally (ROC-AUC = 0.8871, PR-AUC = 0.8611), confirming that learned representations are transferable rather than dataset-specific.

Normal Case (Label = 0.0, Patient ID: 233346)

□ NORMAL CASE | Grad-CAM + Attention | Patient 233346 | Label 0.0



Abnormal Case (Label = 1.0, Patient ID: 237638)

Grad-CAM + MIL Attention | Patient: 237638 | Label: 1.0

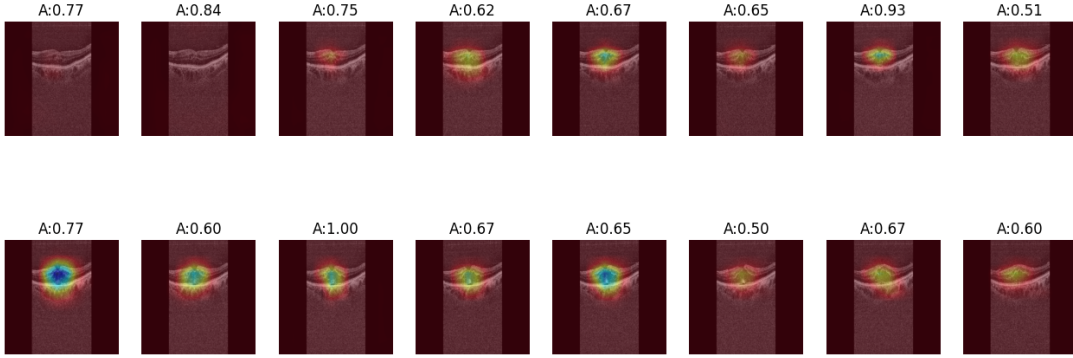


Figure 3.6: Grad-CAM and MIL attention visualizations across OCT slices. Normal cases show diffuse, weak activations, while abnormal cases exhibit localized activations in pathological retinal regions.

Limitations

External evaluation reveals a performance drop relative to internal testing, reflecting residual sensitivity to domain shift from differences in acquisition devices and patient populations. As the model is trained under weak supervision using patient-level labels only, attention maps should not be interpreted as precise lesion segmentations. Finally, while calibration is substantially improved in Model C, post-hoc calibration methods could further enhance probability reliability for clinical deployment.

Dataset	Acc	F1	ROC-AUC	PR-AUC	Sensitivity	Specificity
Internal	0.9688	0.9821	0.9875	–	0.9649	1.0000
External	0.8727	0.7845	0.8871	0.8611	0.7551	0.9247

Table 3.18: Evaluation of the Stage 1 screening model under internal validation using real-world data and external validation using the OCTDL dataset.

Model Selection

Model C consistently outperforms Models A and B across all evaluation dimensions. Its superiority stems from three factors: (1) EfficientNet-B0, providing richer feature representations; (2) gated attention, enabling more discriminative MIL aggregation; and (3) optimized training strategies. Attention, confidence, and Grad-CAM analyses confirm that the learned representations are statistically meaningful and clinically grounded. Model C is therefore selected as the Stage 1 screening component of the proposed framework.

3.3.2 Stage 2: Disease-Specific Multi-Label Classification

Stage 2 performs fine-grained disease-specific multi-label classification on patients with expert-validated pathological labels, using a frozen SimCLR-pretrained ResNet18 encoder with GA-MIL aggregation trained for 30 epochs. The model is evaluated on a held-out test set across three disease categories: DR, AMD, and Edema.

Metric	DR	AMD	Edema	Macro AUC
ROC-AUC	0.90	0.65	0.82	0.79

Table 3.19: ROC-AUC performance of the Stage 2 across different pathological categories on the test set.

The model achieves a macro-average ROC-AUC of 0.79, indicating moderate overall discriminative performance. Edema shows the best separability (ROC-AUC = 0.82), consistent with its clearer OCT signature, while DR also achieves strong performance (ROC-AUC = 0.90). In contrast, AMD is the most challenging class (ROC-AUC = 0.65), reflecting its visual similarity with other early pathological patterns and the difficulty of separating it under weak supervision.

These results provide an initial evaluation of Stage 2 under a frozen encoder setting and

limited expert-labeled data. The observed performance gap for AMD suggests that more expressive MIL formulations, improved multi-label dependency modeling, or fine-tuning of the backbone may be required for more reliable disease-specific classification.

3.4 Conclusion

This chapter presented the experimental evaluation of the proposed multimodal clinical decision support system for retinal disease classification using OCT images and patient metadata. The study was conducted through two complementary frameworks: a controlled ablation study (Framework I) and a real-world two-stage clinical pipeline (Framework II).

Framework I demonstrated that multimodal fusion consistently improves performance over unimodal approaches. In particular, dual cross-attention achieved the best balance between accuracy, calibration, and robustness, while Swin-B provided the most effective image representations among all tested backbones.

Framework II validated the clinical applicability of the system in a hierarchical setting. The screening stage achieved strong and reliable discrimination between normal and abnormal cases, while the second stage provided meaningful disease-specific classification despite inherent inter-class similarities, especially for challenging conditions such as AMD.

Overall, the results confirm that the proposed system is accurate, robust, and clinically relevant, offering a reliable foundation for multimodal AI-assisted retinal disease diagnosis.

Chapter 4

Clinical Decision Support System

4.1 Introduction

This chapter describes the design and implementation of a [Clinical Decision Support System \(CDSS\)](#) that operationalizes the proposed multimodal deep learning framework within a real-world ophthalmological setting. The system assists clinicians in diagnosing macular diseases from [OCT](#) images combined with structured patient metadata.

The primary design objective extends beyond model performance to encompass full end-to-end integration, addressing the practical constraints of clinical deployment: data privacy, operational reliability, and institutional independence. To this end, the system is architected for offline, on-premise deployment within a hospital Local Area Network (LAN), with no dependency on external services or cloud infrastructure.

4.2 System Architecture

The system follows a modular three-tier client–server architecture comprising a presentation layer, an application layer, and a data layer. This separation enforces a clear division of concerns between user interaction, computational logic, and persistent storage, facilitating maintainability and future scalability.

The overall architecture is illustrated in [Figure 4.1](#), and the complete technology stack is summarised in [Table 4.1](#).

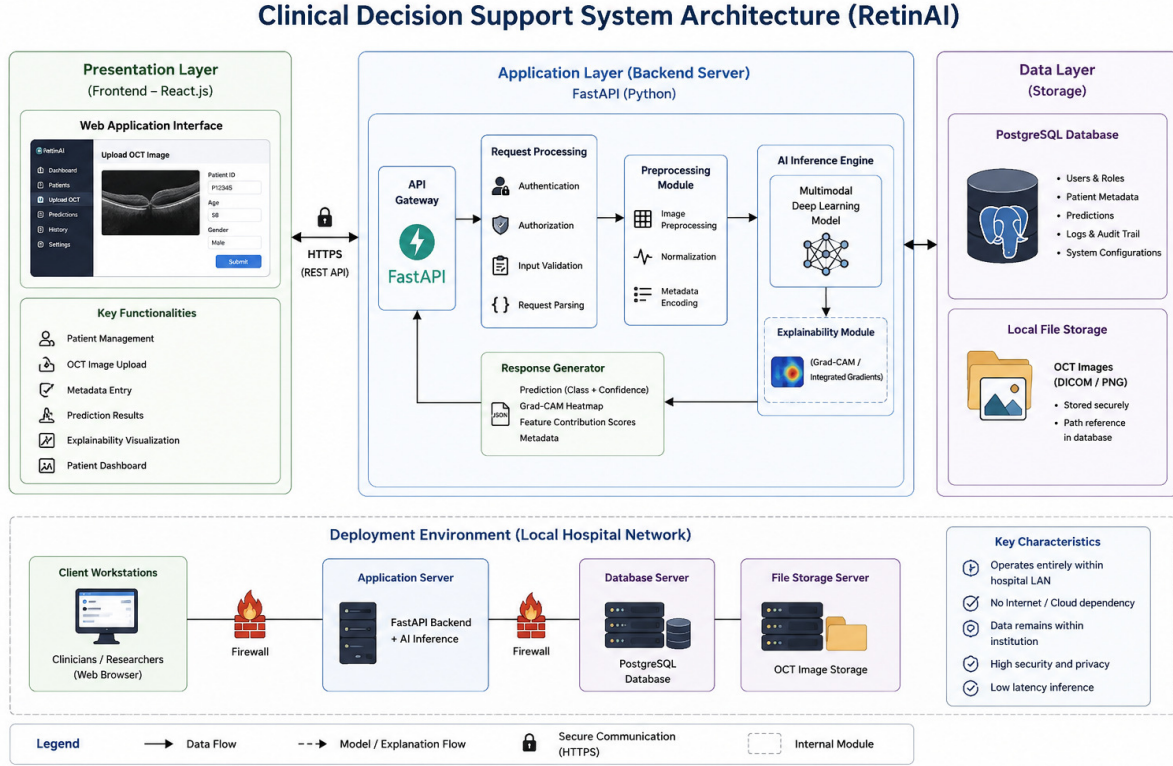


Figure 4.1: Architecture of the proposed clinical decision support system (RetinAI), showing the React frontend, FastAPI backend, AI inference engine, PostgreSQL database, local file storage, and LAN-based deployment environment.

Component	Technology
Frontend Interface	React.js
Backend API	FastAPI (Python)
Database Management	PostgreSQL
Deep Learning Framework	PyTorch
Model Explainability	Grad-CAM, Integrated Gradients
Authentication	JWT-based Role-Based Access Control (RBAC)
Deployment Environment	On-premise LAN (offline clinical deployment)

Table 4.1: Technology stack of the proposed clinical decision support system (RetinAI).

4.2.1 Presentation Layer

The presentation layer is implemented as a React.js single-page application. It exposes a clinician-facing interface for patient record management, [OCT](#) image upload, struc-

tured metadata entry, and interactive visualisation of diagnostic outputs. The interface prioritises responsiveness and clarity to support efficient integration into routine clinical practice.

4.2.2 Application Layer

The application layer is built on FastAPI (Python) and constitutes the computational core of the system. It is responsible for authentication and authorisation, RESTful API request handling, input preprocessing, inference orchestration, and response serialisation.

Specifically, upon receiving a clinician request, the backend (i) validates data completeness and user authorisation, (ii) preprocesses the [OCT](#) image (normalisation) and encodes the structured metadata, (iii) invokes the trained multimodal deep learning model, and (iv) generates structured outputs including the predicted diagnosis, confidence score, [Grad-CAM](#) heatmap, and multimodal contribution indicators. All outputs are serialised as JSON and returned to the frontend for clinical interpretation.

The AI model architecture and training procedure are detailed in the methodology chapter; this layer addresses inference and system integration exclusively.

4.2.3 Data Layer

The data layer comprises a PostgreSQL relational database and a secured local filesystem. The database stores structured patient metadata, model predictions, and system audit logs. [OCT](#) images are stored on the local filesystem, with database-referenced file paths ensuring consistency between imaging data and associated clinical records.

4.3 User Roles and Access Control

The system implements a [Role-Based Access Control \(RBAC\)](#) scheme supported by JWT authentication. Three user roles are defined: **Administrator**, responsible for system configuration, user management, and role assignment; **Clinician**, the primary end user who uploads [OCT](#) scans, enters patient metadata, and interprets AI-generated outputs; and

Researcher, who accesses anonymised datasets for analysis without exposure to identifiable patient information.

This hierarchical structure enforces clinical data governance policies by restricting sensitive operations to authorised users.

4.4 Clinical Workflow

The system workflow is designed to mirror standard ophthalmological diagnostic practice. Following authentication, a clinician creates or selects a patient record and uploads an [OCT](#) B-scan alongside the relevant clinical metadata. The backend validates and preprocesses the input before executing the AI inference pipeline.

The diagnostic output includes the predicted disease class with its associated confidence score, a [Grad-CAM](#) heatmap highlighting diagnostically relevant retinal regions, and multimodal contribution indicators reflecting the relative influence of imaging and metadata features.

The clinician reviews and validates the results directly within the interface. All predictions and user interactions are stored in the database, ensuring traceability, auditability, and reproducibility of diagnostic decisions.

4.5 Data Privacy and Security

The system is designed to comply with clinical data protection regulations by confining all patient data processing to the institutional network. Authentication and [RBAC](#) jointly restrict system access and limit data visibility according to user privileges. Structured clinical data is stored in a secured PostgreSQL instance, while medical images are managed through a controlled local filesystem with restricted access permissions. Together, these mechanisms preserve the confidentiality, integrity, and controlled accessibility of clinical data throughout the system lifecycle.

4.6 Deployment Infrastructure

The system is deployed on a dedicated on-premise server or workstation within the hospital network. The infrastructure comprises a backend node for API processing and AI inference, one or more frontend client workstations, and a local database server. A GPU-enabled machine is recommended to ensure efficient inference, particularly for high-resolution OCT analysis using convolutional or transformer-based architectures. The architecture supports concurrent multi-user access across clinical workstations without modification to the core backend, enabling horizontal scaling across hospital departments while maintaining a centralised computational infrastructure.

4.7 System Demonstration

A complete demonstration of the proposed system was conducted in a simulated clinical environment. The corresponding video is accessible via the QR code presented in Figure 4.2.



Figure 4.2: QR code linking to the RetinAI system demonstration video.

4.8 Conclusion

This chapter presented the design and implementation of a complete clinical decision support system that operationalizes the proposed multimodal deep learning framework within a realistic ophthalmological environment. The system was developed to satisfy key

clinical deployment requirements, including reliability, data privacy, interpretability, and seamless integration into hospital workflows.

A modular three-tier architecture was adopted to separate the presentation, application, and data layers, ensuring maintainability and scalability. The system integrates OCT image processing, structured metadata handling, multimodal inference, and explainability outputs within a unified platform accessible to clinicians through a user-friendly interface.

Deployment on an on-premise infrastructure, combined with role-based access control, ensures secure handling of sensitive patient data while maintaining full institutional control. The inclusion of predictive outputs alongside confidence scores and explainability visualizations enhances transparency and supports clinical interpretability.

Overall, this implementation demonstrates the feasibility of deploying the proposed AI framework as a practical clinical tool, bridging the gap between deep learning research and real-world ophthalmic decision support systems.

Chapter 5

Economic Studies and Strategic Business Planning

5.1 Introduction

The successful deployment of artificial intelligence in clinical practice requires more than high diagnostic accuracy. Beyond technical performance, adoption depends on economic feasibility, workflow integration, infrastructure compatibility, cybersecurity, and long-term sustainability within healthcare environments.

This chapter evaluates the strategic and economic viability of **RetinAI**, an AI-powered clinical decision support platform for **OCT**-based retinal disease diagnosis that integrates multimodal analysis, explainable AI, secure patient management, and role-based clinical workflows.

5.2 Market Analysis and Strategic Positioning

The aim of this section is to validate the commercial and functional viability of RetinAI. It moves beyond technical implementation to demonstrate that the platform addresses a documented gap in the digital healthcare ecosystem and provides a measurable competitive advantage.

5.2.1 Market Definition and Digital Landscape

Market Classification

RetinAI is classified within the following commercial and technical framework:

- **Model Type:** [Business-to-Business \(B2B\) Software-as-a-Service \(SaaS\)](#) AI Healthcare Platform.
- **Market Nature:** Artificial Intelligence in Healthcare, Medical Imaging Analytics, Clinical Decision Support Systems, and Digital Ophthalmology.

The platform targets healthcare institutions involved in retinal imaging and ophthalmic diagnostics, including public hospitals, private ophthalmology clinics, diagnostic imaging centres, university hospitals, telemedicine providers, and medical research institutions. The global AI-in-ophthalmology market is projected to grow at a [Compound Annual Growth Rate \(CAGR\)](#) exceeding 35% through 2030 ([Grand View Research, 2024](#)), driven by aging populations, rising diabetes incidence, and increasing demand for scalable diagnostic imaging.

The Role of AI and Digitalization

RetinAI acts as a *predictive diagnostic engine* to solve the problem of delayed and inconsistent retinal disease interpretation by reducing average [OCT](#) analysis time by approximately **70%**. By leveraging deep learning and multimodal data fusion, the platform transitions ophthalmology departments from manual, specialist-dependent workflows to automated, data-driven clinical decision support.

Traditional retinal diagnosis relies on manual interpretation of [OCT](#) scans by experienced specialists, a process subject to inter-observer variability and difficult to scale in resource-constrained healthcare systems. RetinAI addresses this bottleneck directly through automated disease classification, confidence estimation, and explainable visual outputs.

Scope of Implementation

- **Internal Deployment:** The system operates locally within a hospital [Local Area Network \(LAN\)](#) as a private clinical tool, ensuring full patient data sovereignty, offline capability, low inference latency, and enhanced cybersecurity compliance.
- **External Deployment:** Planned future expansion supports multi-centre hospital networks, cloud-assisted tele-ophthalmology platforms, regional screening programmes, and collaborative research environments.

5.2.2 Target Audience and User Personas

In healthcare technology, a distinction must be drawn between the technical end user and the institutional decision maker.

- **Primary Users:** Ophthalmologists, retinal specialists, imaging technicians, and clinical researchers who interact with the platform daily.
- **Decision Makers:** Hospital directors, healthcare [Information Technology \(IT\)](#) managers, private clinic owners, and public healthcare agency officials who approve technology adoption.
- **User Pain Points:** Target institutions currently face growing imaging volumes, specialist shortages, manual documentation burden, fragmented patient records, and the absence of explainable AI diagnostic tools.
- **Gains:** RetinAI delivers faster diagnosis, reduced clinician workload, improved diagnostic consistency, centralised patient record management, and transparent AI-assisted decision support.

5.2.3 Competitive Benchmarking and Positioning

Direct Competitors

Existing platforms in the ophthalmic AI space include Google Health (retinal AI), ZEISS AI imaging systems, and IDx-DR. Additionally, several academic [OCT](#) classification

frameworks exist, though these lack clinical integration and deployment infrastructure.

Technical Differentiation

Table 5.1 presents a feature-level competitive matrix situating RetinAI relative to the identified competitors across the dimensions of feature completeness, deployment flexibility, and explainability.

Feature	Google Health	ZEISS	IDx-DR / Academic	RetinAI
OCT-based Diagnosis	✓	✓	✓	✓
Clinical Metadata	×	×	×	✓
Patient-Level Reasoning	×	×	×	✓
Explainable AI (full)	Partial	Partial	Limited	✓
Offline Deployment	×	Partial	×	✓
Role-Based	×	Partial	×	✓

Table 5.1: Competitive benchmarking of RetinAI against existing ophthalmic AI solutions.

RetinAI’s primary differentiators are its multimodal OCT-and-metadata fusion, full explainability via Grad-CAM, offline secure deployment, and integrated role-based clinical platform, these features are absent in all benchmarked competitors. The platform is specifically designed to fill the gap between high-accuracy research models and deployable, compliant clinical software.

5.2.4 SWOT Analysis of the Digital Solution

Table 5.2 presents a high-level strategic assessment of RetinAI’s internal capabilities and external environment.

5.3 Business Model and Operational Framework

This section defines the logic by which RetinAI creates, delivers, and captures value. The analysis follows the *Lean Canvas* framework, focusing on the Minimum Viable Product

Strengths (Internal)	Weaknesses (Internal)
Proprietary clinical dataset; multimodal AI architecture; full explainability (Grad-CAM); integrated clinical software platform; offline-first secure deployment.	Limited initial deployment centres; medical device regulatory requirements; GPU infrastructure acquisition cost.
Opportunities (External)	Threats (External)
AI healthcare market growing at CAGR >35%; telemedicine expansion; Algerian and African digital health initiatives; underserved ophthalmic screening demand in emerging markets.	Established international competitors (Google, ZEISS); evolving cybersecurity threats; conservative hospital technology adoption cycles.

Table 5.2: SWOT analysis of RetinAI.

(MVP) and the technical sustainability of the platform.

5.3.1 Value Creation Mechanism

Problem Statement

The following three pain points constitute the primary clinical and operational challenges in the target market:

- **Point 1:** Delayed retinal diagnosis caused by the shortage of available specialist ophthalmologists, resulting in increased disease progression risk.
- **Point 2:** Rising OCT examination volumes that exceed the diagnostic throughput of existing manual workflows.
- **Point 3:** Limited integration of AI diagnostic tools within hospital information systems, leaving clinicians without data-driven decision support.

The AI/Software Solution

RetinAI addresses these challenges through a fully integrated platform offering: AI-powered OCT disease classification, structured patient database management, role-based clinical dashboards for clinicians and researchers, Grad-CAM-based explainability modules, and automated diagnostic report generation.

Minimum Viable Product

To ensure a realistic and deliverable development roadmap, the product scope is divided as follows:

- **Must-Have Features:** User authentication with [RBAC](#), patient record creation and management, [OCT](#) image upload and preprocessing, AI inference pipeline, diagnostic output with confidence score and [Grad-CAM](#) visualisation, structured clinical reporting.
- **Nice-to-Have Features:** Cloud synchronisation, mobile companion application, longitudinal disease progression analytics, multi-language interface support.

Unique Value Proposition

RetinAI is among the first proposed multimodal ophthalmic AI platforms in Algeria, combining patient-level retinal disease diagnosis, full explainable AI, and end-to-end clinical workflow integration within a secure, offline-capable hospital environment

5.3.2 Delivery and Growth Strategy

User Segmentation

RetinAI targets multiple stakeholders within the ophthalmic healthcare ecosystem.

- **Primary Users:** Ophthalmologists, retinal specialists, and imaging technicians who interact directly with the diagnostic platform during routine clinical workflows.
- **Decision Makers:** Hospital directors, clinical managers, and healthcare [IT](#) administrators responsible for evaluating infrastructure compatibility, regulatory compliance, and long-term technology adoption.
- **Research Users:** Academic researchers and medical data scientists interested in anonymized clinical analytics, model evaluation, and future AI development.

Distribution Channels

The initial commercialization strategy focuses on direct deployment within partner hospitals and ophthalmology clinics through pilot installations and institutional licensing agreements.

As the platform matures, distribution will expand to include multi-center hospital deployments, public healthcare partnerships, academic and research collaborations, and API-based integration with electronic medical record systems.

Traction Metrics (KPIs)

To evaluate both technical performance and market adoption, RetinAI monitors the following key performance indicators:

- **Technical Metrics:** AI inference time, diagnostic accuracy, model calibration, and platform uptime.
- **Clinical Metrics:** Number of analyzed [OCT](#) examinations, average diagnostic time reduction, and clinician satisfaction scores.
- **Growth Metrics:** Number of active institutional subscriptions, monthly active users, and retention rate across deployed healthcare institutions.

5.3.3 Economic Sustainability and Cost Structure

Revenue Streams

RetinAI follows a [SaaS](#) licensing model based on annual institutional subscriptions. This model generates predictable recurring revenue while lowering the adoption barrier for healthcare institutions.

Additional long-term revenue opportunities include deployment and installation fees, premium technical support contracts, API licensing for third-party software integration, and advanced analytics modules for research institutions.

Technical Cost Structure

The economic structure of RetinAI includes both capital and operational expenditures.

- **Capital Expenditure (CAPEX):** AI model development, backend and frontend engineering, GPU workstations, security implementation, and regulatory preparation.
- **Operating Expenditure (OPEX):** Local server maintenance, software updates, model retraining, database backup, cybersecurity auditing, and clinical technical support.

Unfair Advantage

RetinAI's long-term competitive sustainability is protected by multiple strategic advantages. First, the platform is built upon a proprietary clinically validated OCT dataset collected from real hospital environments, creating a strong data barrier against competitors. Second, RetinAI combines two independently validated AI frameworks, including multimodal transformer-based diagnosis and patient-level weakly supervised learning. Finally, its first-mover positioning in the Algerian and North African digital ophthalmology market provides a durable regional advantage.

5.3.4 Lean Canvas Summary

Figure 5.1 summarizes the complete business logic of RetinAI across all Lean Canvas dimensions, including problem definition, customer segmentation, value proposition, operational costs, revenue generation, and long-term competitive positioning.

The purpose of this section is to quantify the economic viability of the RetinAI platform and identify potential threats to its execution, including the investment required for development, the break-even threshold, and a structured risk mitigation plan.

5.3.5 Financial Feasibility Analysis

Cost Structure (Investment vs. Operations)

Costs are categorised into the initial build phase CAPEX and the recurring operational phase OPEX:

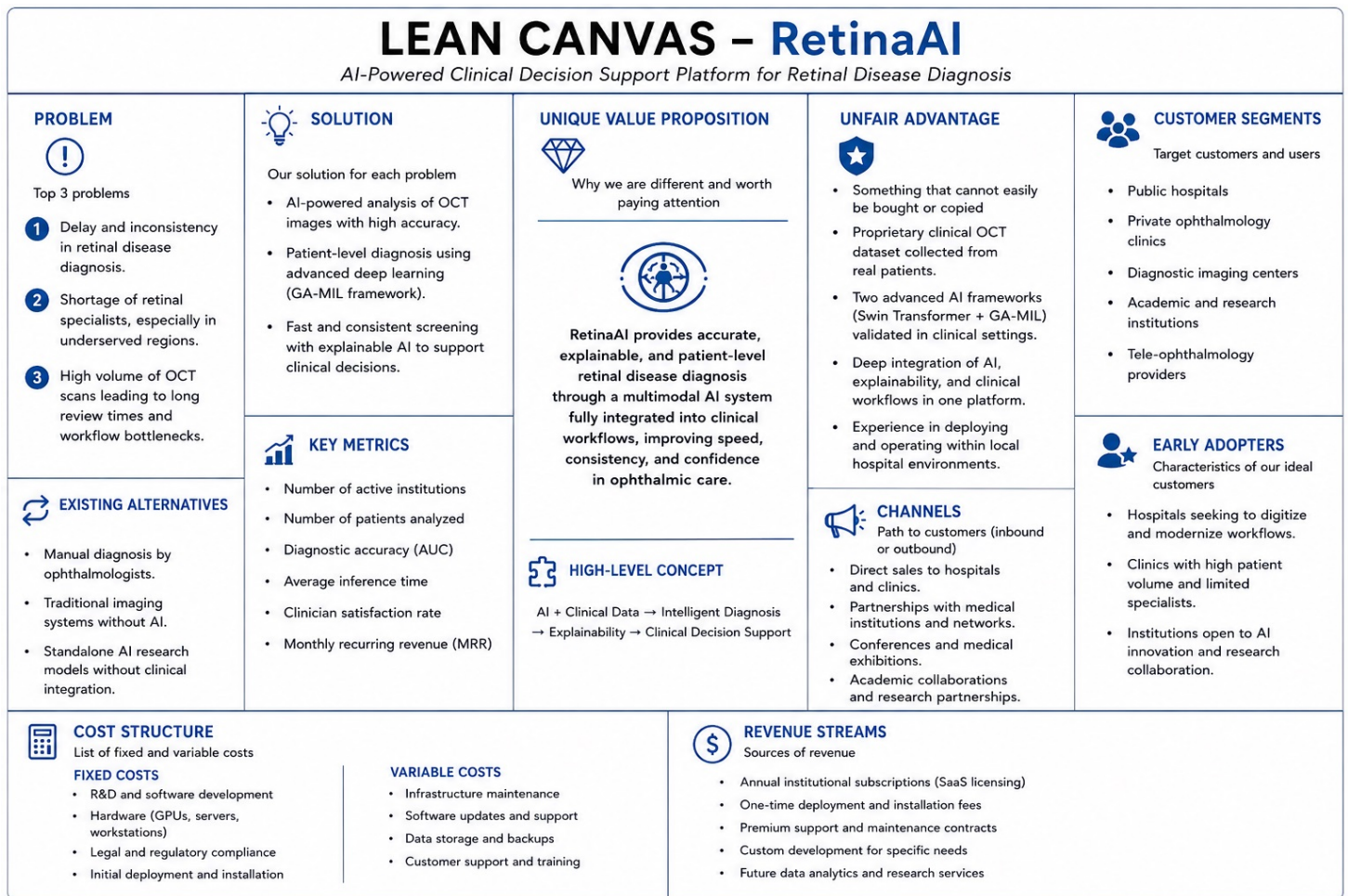


Figure 5.1: Lean Canvas synthesis of the RetinAI digital healthcare startup.

• **Capital Expenditures CAPEX:**

- Development workstations and GPU hardware: 650,000 DZD
- Software engineering and backend development: 800,000 DZD
- Legal protection, IP registration, and deployment testing: 350,000 DZD

$$\text{Total CAPEX} = 1,800,000 \text{ DZD}$$

• **Operating Expenses OPEX:**

- Platform maintenance and software updates: 40,000 DZD/month
- Database management and infrastructure: 35,000 DZD/month

- Security monitoring and technical support: 25,000 DZD/month

$$\text{Total Monthly OPEX} = 100,000 \text{ DZD/month} \quad (1,200,000 \text{ DZD/year})$$

Revenue Model and Projections

RetinAI adopts an annual SaaS subscription model priced at 300,000 DZD per institution per year. Growth assumptions are based on conservative market penetration within the Algerian public and private hospital sector. Assuming deployment across 10 institutions in Year 1, with a projected 20% year-on-year growth in subscriptions:

$$\text{Year 1 Revenue} = 10 \times 300,000 = 3,000,000 \text{ DZD}$$

$$\text{Year 2 Revenue (projected)} = 12 \times 300,000 = 3,600,000 \text{ DZD}$$

This yields a projected net annual surplus of 1,800,000 DZD after OPEX in Year 1, representing a healthy operating margin at modest market penetration.

Break-Even Analysis

The break-even point is the subscription volume at which total revenue equals total costs. It is calculated using the contribution margin formula:

$$N_{\text{break-even}} = \frac{\text{Total Fixed Costs (CAPEX)}}{\text{Revenue per client} - \text{Variable Cost per client}} = \frac{1,800,000}{300,000 - 120,000} = 10 \text{ institutions} \quad (5.1)$$

As shown in Equation 5.1, RetinAI reaches financial break-even upon securing 10 institutional subscriptions. A target considered achievable within the first operational year given the identified demand.

5.3.6 Risk Management

Risk Identification and Categorisation

- **Technical Risks:** Model performance drift due to data distribution shift; patient data privacy breaches; server downtime affecting clinical workflows.
- **Financial Risks:** Infrastructure costs exceeding early-stage revenue; failure to secure pilot hospital partnerships.
- **Organisational Risks:** Regulatory delays in obtaining medical device approval; slow institutional adoption cycles.

Risk Prioritisation (FMEA / RPN Matrix)

Using the [Failure Mode and Effects Analysis \(FMEA\)](#) method, each risk is assigned a [Risk Priority Number \(RPN\)](#) computed as:

$$\text{RPN} = \text{Gravity (G)} \times \text{Probability (P)} \times \text{Detection (D)} \quad (5.2)$$

Table 5.3 defines the action thresholds corresponding to [RPN](#) ranges.

Level	RPN Range	Action Required
Acceptable	1 – 8	Monitor periodically.
Moderate	9 – 27	Implement standard controls.
Critical	28 – 64	Immediate mitigation or contingency plan required.

Table 5.3: RPN-based risk prioritisation levels.

Mitigation Strategies

Table 5.4 summarizes the identified risks, RPN scores, and mitigation strategies.

The highest-risk item is patient data privacy ([RPN](#) = 36), requiring strict security measures including encryption, [RBAC](#), and regular audits. Regulatory delays ([RPN](#) = 32)

Risk	RPN	Mitigation Strategy
Model performance drift	24	Continuous retraining on updated clinical data.
Patient data privacy breach	36	Encryption, RBAC , offline-first design, and security audits.
Regulatory delays	32	Early engagement with health authorities and compliance planning.
Slow institutional adoption	18	Pilot deployments with measurable clinical outcomes.

Table 5.4: FMEA-based risk register for RetinAI.

are mitigated through early compliance planning and coordination with health authorities.

Conclusion

This chapter demonstrates that RetinAI is both a technically validated AI framework and a commercially viable healthcare platform. Market analysis identifies a clear clinical need, while competitive benchmarking and financial projections support its technical value and sustainability potential. Through its proprietary dataset, secure architecture, multimodal diagnostic capabilities, and scalable [SaaS](#) model, RetinAI establishes a solid foundation for supporting the digital transformation of ophthalmology in Algeria and beyond.

General Conclusion

The present work investigated automated retinal disease diagnosis from Optical Coherence Tomography (OCT) images using deep learning, with a focus on multimodal fusion, interpretability, and clinical robustness. The main objective was to design a clinically reliable decision support system that integrates imaging data with structured patient metadata.

Main Contributions

- **Benchmarking of Deep Learning Architectures:** A systematic evaluation of convolutional and transformer-based models for OCT classification was conducted. Results showed that transformer-based architectures, particularly Swin-B, consistently outperform CNN-based models by effectively capturing both local retinal structures and global contextual dependencies.
- **Metadata Recovery from Corrupted Clinical Attributes:** A dedicated metadata recovery module was introduced to reconstruct corrupted or missing clinical attributes (age, sex, eye laterality) directly from OCT images. Both task-specific and multi-head learning strategies were investigated, with task-specific models achieving superior performance due to reduced task interference and better specialization.
- **Multimodal Fusion Strategies:** Different fusion mechanisms were explored to combine OCT images with clinical metadata. Dual cross-attention achieved the best performance by enabling bidirectional interaction between modalities, leading to improved robustness in visually ambiguous cases.

- **Two-Stage Clinical Decision Support System:** A hierarchical framework was designed, consisting of:
 - Stage 1: Patient-level binary screening using Multiple Instance Learning (MIL).
 - Stage 2: Fine-grained disease-specific classification.

This structure closely reflects real clinical diagnostic workflows.

- **Model Reliability and Interpretability:** Calibration metrics (ECE, Brier score) confirmed well-calibrated predictions, while Grad-CAM and attention analyses demonstrated that the model focuses on clinically relevant retinal regions.
- **External Validation and Generalization:** Evaluation on an independent dataset confirmed strong generalization capability, despite observable domain shift across clinical settings.

Limitations

- **Domain Shift and Dataset Bias:** Performance decreases under external validation due to differences in imaging devices, acquisition protocols, and patient distributions, highlighting limited cross-domain robustness.
- **Class Imbalance and Rare Pathologies:** Despite mitigation strategies, performance remains lower for rare classes, reflecting the challenge of learning under severe data imbalance.
- **Limited Metadata Representation:** The available clinical metadata is low-dimensional, limiting the full potential of multimodal fusion and restricting richer clinical context integration.
- **Weak Supervision Constraints:** Both frameworks rely on patient-level labels without lesion-level annotations, restricting fine-grained localization and limiting interpretability to approximate attention maps.

- **Dependence on Data Quality:** Missing slices, noisy labels, or incomplete metadata can negatively impact both stages of the pipeline and propagate errors through the system.
- **Limited External Validation Scope:** External evaluation is conducted on a single dataset, which does not fully capture variability across multiple hospitals or imaging devices.
- **MIL Sensitivity to Slice Selection:** The performance of the screening stage depends on slice selection strategy, where suboptimal sampling may reduce diagnostic reliability.
- **Interpretability Limitations:** Post-hoc methods such as Grad-CAM provide useful visual insights but do not guarantee causal explanations or precise clinical segmentation.
- **Computational Constraints:** Transformer-based multimodal models require significant computational resources, which may limit real-time deployment in low-resource clinical environments.

Future Work

- Improve domain generalization using domain adaptation and multi-center training.
- Leverage self-supervised learning on large-scale unlabeled OCT datasets.
- Incorporate richer clinical data such as medical history and longitudinal records.
- Extend the framework to multi-task learning combining classification and lesion segmentation.

Final Conclusion

This work demonstrates that combining transformer-based vision models with multimodal fusion and attention-based Multiple Instance Learning enables the development of accu-

rate, robust, and clinically reliable OCT-based diagnostic systems. Two complementary frameworks were explored: a multimodal diagnostic model integrating imaging data with structured clinical metadata, and a weakly supervised Multiple Instance Learning pipeline designed for patient-level screening and fine-grained diagnosis.

Across both frameworks, the results highlight the importance of learning rich visual representations while effectively incorporating clinical context when available. The multimodal approach improves performance by reducing ambiguity in visually similar cases, while the MIL-based pipeline provides a clinically aligned hierarchical workflow suitable for real-world screening scenarios.

Beyond performance, this work emphasizes interpretability and reliability. Attention mechanisms and Grad-CAM visualizations help identify relevant retinal structures, while calibration analysis ensures that predicted probabilities remain suitable for clinical decision-making.

Overall, the proposed approaches demonstrate the potential of transformer-based and MIL-based methods as complementary components for retinal disease diagnosis, contributing toward more robust, interpretable, and clinically applicable medical artificial intelligence.

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